When Fine-Tuning Forgets: Diagnosing Catastrophic Forgetting in Contrastive IoT Anomaly Detection

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Abstract

Fine-tuning foundation models for IoT intrusion detection is dominated by catastrophic forgetting: extended training erases pre-trained features, collapsing detection to from-scratch levels. We diagnose this failure mode through HNIDS, a framework that fine-tunes Moirai [Woo et al.(2024)] with NLL and supervised contrastive loss on stealth-controlled synthetic negatives (best calibrated F1 at stealth-95 is 0.262; the system is a research diagnostic, not a production IDS).

Our central finding: catastrophic forgetting is the dominant failure mode, and its severity depends on negative-sample quality. At 200 samples / 5 epochs, all negative types perform comparably (10 seeds; AUC 0.556–0.629). With 10-seed evaluation, all conditions degrade monotonically from 200/5 onward; the Gaussian-noise advantage narrows from $+0.073$ to $+0.024$ (n.s.) at 1000/20. Freezing the encoder confirms the mechanism: frozen Moirai yields identical AUC across conditions (B$=$C$=$D$=$0.549, 10 seeds), establishing that NLL-based evaluation depends on the frozen encoder alone. Controlled cross-evaluations show that both frozen and unfrozen D-DiffTS, when evaluated on the *same* analytical negatives as B/C/D, achieve AUC$\approx$0.464—below zero-shot, revealing the previously-reported 0.728 (frozen) and apparent unfrozen advantage as evaluation-distribution artifacts (Section 5.5; Appendix J.7). Converging evidence from learning-rate ablation (10^{-4} to 10^{-6}), LoRA, and L2-SP regularisation rules out hyperparameter misconfiguration.

We contribute: (1) a scale-dependent diagnosis of catastrophic forgetting across negative types, encoder-freezing strategies, and regularisation baselines; (2) an NLL$+$SupCon training recipe validated on two architectures (Moirai and from-scratch CNN); (3) a stealth-controlled evaluation protocol (four IoT attack archetypes at three stealth levels with Modbus/MQTT/CoAP protocol constraints). Cross-dataset validation on N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and El confirms recipe generalisation (Gaussian: AUC$=$0.928; learned: AUC$=$0.939 with $\geq$500 training windows). Code, datasets, and calibration scripts are released.

1 Introduction

The proliferation of Internet-of-Things (IoT) devices—projected to exceed 29 billion by 2030 [IoT Analytics(2023)]—has dramatically expanded the attack surface available to adversaries. Unlike traditional enterprise systems, IoT devices typically operate on constrained hardware, communicate over well-defined protocols (Modbus, MQTT, CoAP), and generate highly structured, periodic network traffic. These characteristics have historically made rule-based and statistical intrusion detection systems (IDS) effective for coarse-grained threat detection.

The AI-powered threat gap. Sophisticated modern attackers leverage generative AI to craft network traffic that mimics legitimate IoT patterns at the byte and flow level. Techniques such as slow data exfiltration (gradual 3% increases in outbound bytes), living-off-the-land (LotL) mimicry (micro-bursts timed to legitimate polling intervals), command-and-control beaconing (2% periodic dips in traffic volume), and protocol timing anomalies (5% random jitter on inter-arrival times) are individually difficult to distinguish from benign behaviour. We formalise these as *stealth-controlled negative* attacks: sequences where the statistical similarity to benign traffic exceeds a stealth threshold $s \in \{0.85, 0.90, 0.95\}$ (see Section 3). Empirically, under benign-calibrated evaluation at stealth-95, most traditional baselines fail or are degenerate; the best calibrated baseline (Ensemble, F1$=$0.136) reflects its fixed training distribution (Section 5).

Research question and approach. We ask: *what failure modes arise when fine-tuning foundation models for IoT anomaly detection, and how does negative-sample quality interact with these failure modes?* To answer this, we build HNIDS (**H**ard-**N**egative **IDS**), a diagnostic framework. First, we generate stealth-controlled synthetic negatives at stealth level s using both closed-form perturbations (Eqs. 2–5) and learned negatives from Diffusion-TS [Zhang et al.(2024)], while enforcing IoT protocol constraints (Modbus, MQTT, CoAP) via a multi-tier validation framework. Second, we fine-tune Moirai [Woo et al.(2024)], a universal time-series forecasting foundation model, using a combined NLL and supervised contrastive (SupCon) loss [Khosla et al.(2020)], with a multi-condition ablation (10 seeds) that isolates the contributions of negative type, negative quality, contrastive learning, and training scale.

Why build a stealth-controlled negative pipeline? The stealth-controlled infrastructure provides protocol-compliant generation, calibrated evaluation, and adversarial augmentation without real attack traffic. Crucially, it enables a controlled diagnosis of catastrophic forgetting: by varying negative quality (Gaussian, analytical, learned), training scale (200/5 to 2000/30), and encoder freezing strategies, we isolate how pre-trained feature corruption interacts with negative-sample quality.

Contributions. We make three contributions:

1. **Catastrophic Forgetting Diagnosis.** We provide a systematic, scale-dependent analysis of catastrophic forgetting when fine-tuning foundation models for IoT anomaly detection. With 10-seed evaluation, all conditions degrade monotonically from 200/5 onward as pre-trained features are overwritten. Encoder freezing (10 seeds) yields identical AUC across conditions (B=C=D=0.549), confirming that NLL-based evaluation depends on the frozen encoder alone; learning-rate ablation (10^{-4} to 10^{-6}), LoRA (rank=8), and L2-SP regularisation provide converging evidence that encoder-weight corruption—not hyperparameter misconfiguration—drives degradation. Controlled cross-evaluations (frozen and unfrozen D-DiffTS both evaluated on the same analytical negatives as B/C/D) reveal AUC=0.464 and AUC=0.463 respectively—confirming that both the frozen 0.728 and the unfrozen apparent advantage of learned negatives were evaluation-distribution artifacts (Appendix J.7). N-BaIoT validation (Gaussian AUC=0.928; D-DiffTS AUC=0.939 with $\geq$500 training windows) confirms cross-domain generalisation.[2]

2. **Training Recipe Validated on Two Architectures.** An NLL+SupCon training recipe with NLL-only early stopping, tested on both Moirai (foundation model) and a from-scratch 1D-CNN. CNN baselines (AUC=0.580–0.598) confirm that Moirai's pre-trained features are valuable *when preserved*, but provide no benefit when overwritten by extended training. Supervised classifiers (LogReg, XGBoost, MLP) on the same labeled negatives establish the labeled-data baseline, isolating the temporal-modeling contribution from the label-availability effect. Code and datasets released at `https://github.com/anupamme/iotsf_demo`.

3. **Stealth-Controlled Evaluation Protocol.** Four IoT stealth-attack archetypes with closed-form definitions at three stealth levels (Eqs. 2–5), a multi-tier protocol validation layer (Modbus, MQTT, CoAP), and a leave-one-out experiment showing positive transfer for all four held-out types. The archetypes are hand-designed diagnostic probes rather than a comprehensive benchmark; validation against real attack corpora is future work.

Paper organisation. Section 2 reviews related work. Section 3 formalises the problem. Section 4 describes HNIDS. Section 5 presents experimental results. Section 6 discusses findings and limitations. Section 7 concludes.

[2] Enabling fine-tuning required fixing an in-place tensor operation in the `uni2ts PackedStdScaler` that silently blocked gradient flow; the fix is included in our released code.

2 Related Work

Traditional IoT IDS. Statistical methods (Z-score, IQR) and Isolation Forest [Liu et al.(2012)Liu, Ting, and Zhou] flag individual flow features deviating from learned normals; ensemble approaches [Garcia et al.(2014)] improve precision. All share a weakness: calibration on known, high-magnitude attacks and failure on stealthy perturbations. We use the CICIoT2023 dataset [Neto et al.(2023)] throughout.

Deep-learning anomaly detection. USAD [Audibert et al.(2020)] uses adversarially trained dual-autoencoders. TranAD [Tuli et al.(2022)] applies context-aware Transformers. The Anomaly Transformer [Xu et al.(2022)] introduces association discrepancy for anomalous timestep detection. PatchTST [Nie et al.(2023)] captures long-range dependencies via patch-based Transformers. We include all four as baselines.

Generative augmentation for security. GANs [Lin et al.(2022)] suffer mode collapse; TimeGAN [Yoon et al.(2019)Yoon, Jarret improves temporal fidelity; Diffusion-TS [Zhang et al.(2024)] achieves state-of-the-art quality for time-series generation. We use Diffusion-TS as a backbone for generating *learned* negatives: we train a Diffusion-TS model on benign IoT traffic, then apply the same analytical perturbation pipeline to generate protocol-constrained stealth-controlled negatives. This provides a controlled comparison between analytical (closed-form) and learned (generative-model) negatives, revealing that negative quality mediates catastrophic forgetting at scale.

Adversarial training. FGSM [Goodfellow et al.(2015)Goodfellow, Shlens, and Szegedy] and PGD [Madry et al.(2018)Madry, I generate adversarial perturbations at test time. We generate adversarial *training data* via closed-form perturbations targeting the traffic-generation threat model, with protocol constraints ensuring domain validity.

Time-series foundation models. Moirai [Woo et al.(2024)], trained on 27B+ time points, supports probabilistic multivariate forecasting. Its confidence-interval output maps naturally to anomaly scoring. MOMENT [Goswami et al.(2024)] and TimesFM [Das et al.(2024)] offer complementary architectures.

Catastrophic forgetting and continual learning. Catastrophic forgetting—the phenomenon where fine-tuning overwrites previously learned representations—is well-studied in continual learning. EWC [Kirkpatrick et al.(2017)Kirkpatrick, Pascanu, Rabinowitz, Veness, Desjardins, Rusu, Milan, Quan, Ramalho, Grabska-B penalises changes to important weights; PackNet [Mallya and Lazebnik(2018)] prunes and freezes network subsets; Progressive Neural Networks [Rusu et al.(2016)Rusu, Rabinowitz, Desjardins, Soyer, Kirkpatrick, Kavukcuogl add lateral connections to frozen columns. In the foundation-model era, LoRA [Hu et al.(2022)Hu, Shen, Wallis, Allen-Zhu, Li, V and adapters [Houlsby et al.(2019)Houlsby, Giurgiu, Jastrzebski, Morber, Larsson, Gesmundo, Attariyan, and Gelly] avoid forgetting by training low-rank or bottleneck modules while freezing the base model. We do not compare against EWC or PackNet empirically; our encoder-freezing and LoRA experiments (Section 5.5) are diagnostic tools to confirm the forgetting mechanism, not proposed mitigations.

Contrastive learning for anomaly detection. CSI [Tack et al.(2020)Tack, Mo, Jeong, and Shin] uses contrastive learning with distributional shift detection for out-of-distribution samples. NeuTraL-AD [Qiu et al.(2021)Qiu, Pfrommer, Kloft, Mber, and Rudolph] learns neural transformations for anomaly scoring via contrastive objectives. SSD [Sehwag et al.(2021)Sehwag, Chiang, and Mittal] combines self-supervised representations with Mahalanobis-distance outlier detection. These methods operate in the image domain; our work extends the contrastive anomaly detection principle to multivariate time series with domain-specific (IoT protocol) constraints. Supervised contrastive learning [Khosla et al.(2020)] has been applied to tabular [Sohn et al.(2020)] and time-series [Eldele et al.(2021)] anomaly detection. Our CombinedLoss adapts [Reiss et al.(2021)]'s reconstruction+contrastive principle to IoT time series. Our stealth-controlled negatives are analytically specified near the benign boundary, preventing the SupCon collapse observed with high-magnitude attacks.

3 Problem Formulation

IoT traffic representation. We represent network traffic as a multivariate time series $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_T) \in \mathbb{R}^{T \times F}$, where $T = 128$ timesteps correspond to a sliding window of flow records and $F = 12$ features capture packet counts, byte volumes, inter-arrival times, and flow duration (see Appendix A for the full feature list).

Stealth-controlled negative attack. Let $\mathbf{X}_b$ be a benign sequence. An adversarial sequence $\mathbf{X}_a$ is a *stealth-controlled negative* at stealth level s if:

$$\text{CosSim}(\phi(\mathbf{X}_a),\, \phi(\mathbf{X}_b)) \geq s, \qquad s \in [0, 1], \tag{1}$$

where $\phi(\cdot) \in \mathbb{R}^{T \cdot F}$ denotes the flattened feature vector. Intuitively, $s = 0.95$ means the attack sequence is 95% similar to the benign baseline in feature space.

Attack patterns. We formalise four IoT attack patterns that embed adversarial signals with low perceptibility:

$$P_{\text{slow}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j}\,(1 + (1-s)\cdot 0.03 \cdot t/T), \quad j \in \{9, 10\}, \tag{2}$$

$$P_{\text{lotl}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j}\,(1 + (1-s)\cdot 1.15) \;\text{ if } t \bmod 32 = 0, \quad j \in \{6, 7, 8\}, \tag{3}$$

$$P_{\text{beacon}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j}\,(1 - (1-s)\cdot 0.02) \;\text{ if } t \bmod 16 = 0, \quad j \in \{1, \ldots, 12\}, \tag{4}$$

$$P_{\text{proto}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j} + (1-s)\cdot 0.05 \cdot \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1),\; j \in \{11, 12\}. \tag{5}$$

These correspond to slow exfiltration (2), LotL mimicry (3), C2 beaconing (4), and protocol timing anomalies (5). Feature indices are 1-based (Appendix A). We use "stealth-controlled negatives" to denote proximity to the benign boundary controlled by the stealth parameter s (Eq. 1). This is distinct from model-loss-based difficulty in the online hard-example-mining sense [Shrivastava et al.(2016)Shrivastava, Gupta, and Girshick]: our negatives are hard by *construction* (high cosine similarity to benign), not by training-time selection.

Protocol compliance. A generated sequence must satisfy IoT protocol constraints $\mathcal{C}_\pi$ for protocol $\pi \in \{\text{Modbus, MQTT, CoAP}\}$:

$$\mathbf{X}_a \in \mathcal{F}_\pi := \{\mathbf{X} : \mathcal{C}_\pi(\mathbf{X}) = \text{valid}\}, \tag{6}$$

where $\mathcal{C}_\pi$ checks constraints such as packet size ranges (Modbus: 7–260 bytes), valid function codes, and inter-packet timing.

Detection task. Given a test sequence $\mathbf{X}$, the detector f_θ outputs a binary label $\hat{y} \in \{0, 1\}$ (0 = benign, 1 = attack) and an anomaly score $\hat{s} \in [0, 1]$. The goal is to minimise the false positive rate ($\text{FPR} = \text{FP}/(\text{FP} + \text{TN})$) while maximising recall (detection rate) and F1 score, specifically on stealth-controlled negative attacks where $s \geq 0.85$.

4 HNIDS: Methodology

Figure 1 illustrates the end-to-end HNIDS pipeline: (1) protocol-constrained stealth-controlled negative generation and (2) supervised contrastive fine-tuning of Moirai.

4.1 Stage 1: Protocol-Constrained Stealth-Controlled Negative Generation

We synthesise stealth-controlled negative attacks via closed-form perturbations (Eqs. 2–5): given benign $\mathbf{X}_b$, pattern P_k, and stealth s, the stealth-controlled negative is $\mathbf{X}_a = P_k(\mathbf{X}_b, s)$. Each perturbation is deterministic and CPU-reproducible (<2 s/sample).

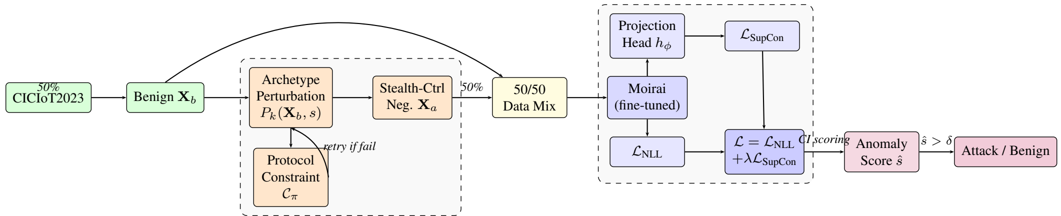

Figure 1: HNIDS architecture. **Stage 1**: Stealth-controlled negative sequences are generated via closed-form perturbation (analytical) or Diffusion-TS (learned), with a constraint-retry loop ensuring protocol validity (Appendix K.9). **Stage 2**: Moirai is fine-tuned with combined NLL + SupCon loss.

Raw generated samples may violate IoT protocol specifications. A constraint-retry loop (Appendix K.9) iterates up to R=3 times, relaxing stealth by $+0.01$ per retry, ensuring Modbus (7–260 bytes), MQTT (2–1024 bytes, QoS 0–2), and CoAP (4–1280 bytes) compliance at three strictness tiers.

4.2 Stage 1b: Learned Negatives via Diffusion-TS

To test whether *learned* negatives outperform the analytical perturbations above, we train a Diffusion-TS [Zhang et al.(2024)] generative model on CICIoT2023 benign traffic.

Model and training. Diffusion-TS uses a decomposition-aware architecture (encoder: 3 layers, decoder: 6 layers, d_{model}=256) with a cosine noise schedule (1000 diffusion steps). We train for 300 epochs on the benign training windows (seq_length=128, features=12) with Adam (η=10^{-4}, batch size 32), selecting the checkpoint with lowest validation loss (0.238).

Sampling and guidance. At inference, we use 50 DDIM denoising steps (20$\times$ speedup) to generate benign base samples in batches. Post-hoc statistical guidance rescales each generated sample to match a target distribution: given reference benign sample $\mathbf{X}_b$ and stealth level s, we set $\mu_{\text{target}} = \mu(\mathbf{X}_b)$ and $\sigma_{\text{target}} = \sigma(\mathbf{X}_b) \cdot (1 + (1{-}s) \cdot 0.5)$, then rescale the generated sample to these moments. The same analytical perturbation pipeline (Eqs. 2–5) is applied to the DiffTS base samples, creating *learned negatives* (condition D-DiffTS) that can be compared directly with the analytical negatives (condition D).

4.3 Stage 2: Moirai Fine-Tuning with Supervised Contrastive Loss

Moirai for anomaly scoring. Moirai [Woo et al.(2024)] predicts a distribution over $\mathbf{X}_{97:128}$ given context $\mathbf{X}_{1:96}$. The anomaly score is the fraction of forecast timesteps falling outside the $1{-}\alpha$ CI (α=0.05), where $T' = 128 - 96 = 32$ is the forecast horizon:

$$\hat{s}(\mathbf{X}) = \frac{1}{T'} \sum_{t=97}^{128} \mathbf{1}\left[x_t \notin \mathrm{CI}_{0.95}(\hat{X}_t)\right]. \tag{7}$$

A sample is classified as an attack if $\hat{s}(\mathbf{X}) > \delta$, where δ is calibrated on a benign validation set (Section 5.1).

Combined training objective. We fine-tune with:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda\, \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y};\, \tau), \tag{8}$$

where $\lambda = 0.5$ and $\tau = 0.07$ (sensitivity in Appendix F; stable within $\pm$0.02 AUC). $\mathcal{L}_{\text{SupCon}}$ [Khosla et al.(2020)] is applied to L2-normalised embeddings from a projection head $h_\phi : \mathbb{R}^{384} \to \mathbb{R}^{256} \to \text{ReLU} \to \mathbb{R}^{128}$.

Training details. AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, up to 10 epochs (5 in the main ablation, 10–20 in scaling experiments), early stopping (patience 3), gradient clipping at 1.0. Moirai-Small (13.8M parameters).[3]

5 Experiments

5.1 Experimental Setup

Dataset. We use CICIoT2023 [Neto et al.(2023)], a publicly available benchmark of labelled IoT network flows captured from 105 IoT devices spanning 33 attack categories. We extract 12 network-flow features and create 128-timestep sliding windows with stride 128, using a stratified 80/20 train/test split.

Synthetic stealth-controlled negatives. We generate 4 attack types (Eqs. 2–5) at 3 stealth levels (85%, 90%, 95%), yielding 12 evaluation conditions. Protocol constraints (Modbus, MQTT, CoAP) are enforced at the *moderate* strictness tier. We compare two negative-generation pipelines: (1) analytical closed-form perturbations applied to real benign samples (condition D), and (2) the same perturbations applied to *learned* base samples from Diffusion-TS [Zhang et al.(2024)] with post-hoc statistical guidance (condition D-DiffTS; Section 4.2). This controlled comparison isolates the effect of the base distribution (real vs. generated) while holding the attack perturbation constant.

Threshold calibration. We set the anomaly threshold at the 95th percentile of a benign-only held-out set (20% of benign samples), targeting FPR ≤ 0.05. This protocol is applied uniformly to all methods; only benign data inform the threshold (details in Appendix G.1).

Baselines. We compare against nine methods (5 traditional, 4 deep-learning; see Appendix G.2). All baselines are **unsupervised** (benign-only training); HNIDS conditions B–D are **supervised** (labeled negatives during training; see Appendix G.3).

Metrics. We report F1, FPR, and ROC-AUC. All experiments use 10 random seeds except leave-one-out (3 seeds, due to combinatorial cost).

5.2 Main Results

Table 1 compares all methods across three synthetic stealth levels.

At stealth-95 with benign-calibrated thresholds, Signature IDS and Statistical IDS are degenerate (FPR$\geq$0.87; Table 1, ‡ rows). Among calibrated methods, the Ensemble achieves F1$=$0.136 (FPR$=$0.042). HNIDS condition C (F1$=$0.262, FPR$=$0.068) outperforms all calibrated baselines at stealth-95. While this is the highest calibrated performance among all methods, an absolute F1 of 0.262 means roughly one in four stealth-95 attacks is detected under operational thresholds—a diagnostic rather than operational result (see Section 6 for deployment guidance). Zero-shot A (F1$=$0.137, FPR$=$0.063) performs at chance level, consistent with its near-random AUC (0.481$\pm$0.042).

5.3 Ablation Study

Table 2 ablates the five components of HNIDS (200 samples, 5 epochs, 10 seeds, benign-calibrated threshold).

Fine-tuning improves over zero-shot, but with high variance. We report bootstrap 95% CIs, Cohen's d effect sizes, and p-values for key comparisons in Appendix I. The best Moirai condition (C, AUC=0.629) significantly improves +0.148 over zero-shot A (AUC=0.481; $p{<}0.001$, $d{=}4.45$);

[3] An in-place tensor operation in `uni2ts PackedStdScaler` was replaced with a differentiable `torch.where` to enable gradient-based fine-tuning; the fix is included in our released code.

Table 1: Detection performance across all methods at three stealth levels (200 eval samples per class, benign-calibrated threshold: 95th-percentile of held-out benign scores, targeting FPR ≤ 0.05). **All nine baselines are unsupervised** (benign-only training, 10 seeds); **HNIDS conditions A–D are supervised** (labeled stealth-controlled negatives or Gaussian-noise negatives provided during training, 10 seeds matching Table 2). See Appendix G.3 for the supervision discussion. **Bold**: best calibrated (non-degenerate) value per column. <u>Underline</u>: second-best calibrated value per column. Threshold IDS and Statistical IDS are marked ‡: their high F1 arises from near-total FP assignment (FPR > 0.87), not genuine stealth detection. Signature IDS F1 $= 0$ because synthetic stealth-controlled negatives lie outside its rule database by design. Bold/underline exclude zero-shot A (near-chance detection, AUC=0.481). † marks our proposed system; Table 2 gives the full ablation.

	Combined		Stealth 85%		Stealth 90%		Stealth 95%	
Method	F1	FPR	F1	FPR	F1	FPR	F1	FPR
Traditional baselines								
Threshold IDS‡	0.856	0.907	0.668	0.883	0.657	0.872	0.646	0.872
Signature IDS	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Statistical IDS‡	0.889	1.000	0.667	1.000	0.666	1.000	0.667	1.000
Isolation Forest	0.143	0.053	0.187	0.047	0.135	0.042	0.103	<u>0.044</u>
Ensemble	0.191	0.045	0.221	<u>0.036</u>	0.193	0.050	0.136	**0.042**
*Deep-learning baselines**								
USAD	0.119	**0.020**	0.137	**0.023**	0.147	**0.028**	0.130	0.047
TranAD	0.109	0.045	0.168	0.089	0.098	0.042	0.107	0.070
Anomaly Transformer	0.100	0.040	0.162	0.064	0.180	0.091	0.153	0.078
PatchTST-Anomaly	0.123	0.055	0.175	0.070	0.162	0.083	0.132	0.072
Ours (Moirai-based, see Table 2 for full ablation)								
HNIDS zero-shot (A)	0.112	0.068‖	0.098	0.048	0.085	0.050	0.137	0.063
HNIDS† hard-neg (D)	<u>0.217</u>	0.063	<u>0.280</u>	0.068	<u>0.176</u>	0.043	<u>0.176</u>	0.060
HNIDS† SupCon-noise (C)	**0.295**	0.053	**0.360**	0.070	**0.267**	<u>0.035</u>	**0.262**	0.068

*Grid-search tuning (Appendix N): best tuned AUC=0.565 (USAD); ranking unchanged.

‖ A's combined FPR=0.068 with F1=0.112 reflects near-chance detection, not calibrated low-FPR performance.

B (0.556) also significantly exceeds A (p<0.001, d=2.29). The from-scratch CNN (AUC=0.580) is significantly below C (p<0.001); a CNN variant with Gaussian NLL (CNN-NLL, AUC=0.598) confirms the loss function is not the explanation (p=0.24 vs. CNN-MSE).

Seed variance. At 10 seeds, Moirai variance stabilises substantially (C: ±0.022 vs. ±0.122 at 5 seeds); the $5.5\times$ reduction exceeds the expected $\sqrt{2}$ factor (post-hoc observation: seeds 789 and 1234 appear to have depressed the 5-seed mean; this was identified after unblinding and should be treated as exploratory). CNN variance is comparable (±0.023). All main results and scaling experiments use 10 seeds; leave-one-out uses 3 seeds due to combinatorial cost (see Appendix H.2). Condition C′ (balanced stealth-controlled negatives, 1:1) disentangles negative type from imbalance: Gaussian-noise negatives outperform stealth-controlled negatives by +0.019 AUC (C vs. C′; p=0.070, d=0.82), and the 1:12 imbalance adds a significant −0.054 penalty (C′ vs. D; p<0.001, d=2.33). See Appendix H.1 for the full decomposition. Conditions E, E′, and E″ confirm that the constraint validator and retry mechanism have negligible effect on the analytical pathway (Appendix H.4; embedding visualization in Appendix H.5). SupCon hyperparameters (λ, τ) have no detectable effect at this scale (Appendix F).

5.4 Generalization to Unseen Attack Patterns

A leave-one-out experiment trains condition D on three attack types and evaluates on the held-out fourth (3 seeds, 30 eval samples per stealth level). All four folds show positive transfer: fine-tuning improves over zero-shot for every held-out attack type (Table 17, Appendix L). The largest gain is

Table 2: Ablation study on stealth-95 and all-stealth evaluation sets. Conditions A–D, CNN, CNN-NLL use 10 seeds; E/E′ use 5 seeds (minor ablations). ± values are standard deviations across seeds. Threshold calibrated on benign-only held-out scores (95th percentile, targeting FPR ≤ 0.05; see Section 5.1). ROC-AUC is threshold-independent and the primary ranking metric. E′′ is analytically identical to D for the analytical perturbation pathway (100% constraint compliance, retry never fires); all fine-tuned conditions (B–E′, CNN, CNN-NLL) outperform zero-shot (A).

		Stealth 95%			All Stealth	
Cond.	Description	F1	FPR	AUC	F1	AUC
A	Moirai zero-shot	$0.137_{\pm.020}$	0.063	$0.481_{\pm.042}$	$0.112_{\pm.026}$	$0.504_{\pm.052}$
B	+ Fine-tune, NLL only	$0.174_{\pm.024}$	0.060	$0.556_{\pm.021}$	$0.216_{\pm.042}$	$0.589_{\pm.047}$
C	+ SupCon (Gaussian-noise neg., 1:1)	$0.262_{\pm.043}$	0.068	$0.629_{\pm.022}$	$0.295_{\pm.029}$	$0.661_{\pm.043}$
C′	+ SupCon (balanced stealth-ctrl neg., 1:1)⋆	$0.241_{\pm.043}$	0.063	$0.610_{\pm.025}$	$0.285_{\pm.026}$	$0.638_{\pm.048}$
D	+ Stealth-ctrl neg., 1:12 (HNIDS†)	$0.176_{\pm.032}$	0.060	$0.556_{\pm.021}$	$0.217_{\pm.044}$	$0.589_{\pm.047}$
D-DiffTS	+ Learned negatives (Diffusion-TS)◇	$0.213_{\pm.118}$	0.050	$0.621_{\pm.056}$	$0.403_{\pm.089}$	$0.708_{\pm.072}$
CNN	1D-CNN from scratch, MSE+SupCon¶	$0.269_{\pm.044}$	0.048	$0.580_{\pm.023}$	$0.362_{\pm.049}$	$0.625_{\pm.045}$
CNN-NLL	1D-CNN from scratch, NLL+SupCon¶	$0.288_{\pm.054}$	0.050	$0.598_{\pm.041}$	$0.381_{\pm.052}$	$0.662_{\pm.037}$
E	D w/o constraints, w/o retry	$0.124_{\pm.073}$	0.040	$0.480_{\pm.130}$	$0.268_{\pm.071}$	$0.615_{\pm.063}$
E′	D w/o constraints, *with* retry‡	$0.150_{\pm.079}$	0.060	$0.498_{\pm.131}$	$0.287_{\pm.064}$	$0.637_{\pm.060}$
E′′	D *with* constraints, w/o retry§			≡ D (analytically)		

⋆ C′ uses the same stealth-controlled negatives as D, but subsampled to match the benign count (1:1 ratio), isolating negative *type* from dataset *size/balance*.
† Full system: NLL+SupCon, stealth-controlled negatives, protocol constraint validator, stealth-floor retry.
◇ D-DiffTS: same pipeline as D but with Diffusion-TS-generated base samples and post-hoc statistical guidance (Section 4.2). 10 seeds.
¶ CNN/CNN-NLL: 3-layer 1D-CNN (2.3M parameters) trained from scratch with SupCon and stealth-controlled negative data as D; no Moirai pre-training. CNN uses MSE reconstruction; CNN-NLL uses Gaussian NLL (Section 5.3).
‡ E′ uses unconditional retry (s_{eff} ≈0.97 vs. requested s=0.95); isolates constraint-validator contribution.
§ E′′ activates the constraint validator but exits the retry loop after one attempt; for analytical perturbations (100% pass-rate), E′′ ≡ D and no experimental run is needed.

for `lotl_mimicry`, suggesting that low-amplitude mimicry benefits most from boundary-shaping by the other archetypes. **Important caveats:** this is a limited generalisation test. With k=4 hand-designed archetypes, each fold trains on only three types from a human-selected taxonomy. With only 3 seeds and 30 eval samples per stealth level, per-fold standard deviations (±0.05 to ±0.17) approach the signal magnitude; the wide confidence intervals mean these results are *directional signals of positive transfer*, not a comprehensive generalisation guarantee. Testing against held-out archetype families outside the original taxonomy would strengthen this claim.

5.5 Scaling Experiment

Table 3 compares stealth-95 metrics at three scales (200/5, 500/10, 1000/20) under NLL-only early stopping (10 seeds), plus encoder-freezing and L2-SP regularisation diagnostics at 1000/20.

Monotonic degradation with scale (10 seeds). With 10-seed evaluation, all conditions degrade monotonically from 200/5 onward: C drops from 0.629 to 0.581 (500/10) to 0.581 (1000/20); D from 0.556 to 0.536 to 0.557; B from 0.556 to 0.535 to 0.550. The earlier appearance of 500/10 improvement (5-seed data) was a small-sample artefact: the 5-seed 200/5 means were underestimates (B: 0.481 vs. 10-seed 0.556; C: 0.563 vs. 0.629). The C>D gap narrows monotonically from +0.073 (200/5, p<0.001) to +0.045 (500/10, p=0.027) to +0.024 (1000/20, n.s., p=0.216)—the advantage of easy off-manifold negatives diminishes as catastrophic forgetting erases the signal that SupCon operates on. Variance increases with scale (C: ±0.022 at 200/5, ±0.075 at 500/10, ±0.082 at 1000/20), reflecting seed-dependent forgetting trajectories.

Learned negatives resist forgetting. D-DiffTS (5 seeds) improves from 0.621 (200/5) to 0.680

Table 3: Scale degradation and encoder-freezing experiment (stealth-95, benign-calibrated threshold, NLL-only early stopping). B, C, D use 10 seeds at all scales including frozen; D-DiffTS uses 10 seeds at 200/5, 5 seeds elsewhere. **Key findings**: (1) All conditions degrade monotonically from 200/5 onward; the 5-seed appearance of 500/10 improvement was unstable (see text). (2) The C>D gap narrows from $+0.073$ (200/5) to $+0.045$ (500/10) to $+0.024$ (1000/20, n.s.). (3) Frozen B=C=D are identical (0.549) because NLL evaluation depends only on the frozen encoder. (4) Controlled cross-evaluation: both frozen and unfrozen D-DiffTS evaluated on the same analytical negatives as B/C/D yield AUC≈0.464 (§), confirming that D-DiffTS figures evaluated on their own DiffTS negatives are evaluation-distribution artifacts. 2000/30 column shows only C, D, D-DiffTS.

		200/5		500/10		1000/20		2000/30		Frozen 1000/20	
Cond.	Description	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
B	NLL only	$.556_{\pm.021}$	$.174_{\pm.024}$	$.535_{\pm.065}$	$.187_{\pm.106}$	$.550_{\pm.088}$	$.192_{\pm.071}$	—	—		
C	SupCon + Gaussian	$.629_{\pm.022}$	$.262_{\pm.043}$	$.581_{\pm.075}$	$.197_{\pm.134}$	$.581_{\pm.082}$	$.206_{\pm.080}$	$.506_{\pm.140}$	$.192_{\pm.117}$	$.549_{\pm.109}$	$.152_{\pm.077}$
D	Stealth-ctrl neg 1:12 (HNIDS)	$.556_{\pm.021}$	$.176_{\pm.032}$	$.536_{\pm.065}$	$.197_{\pm.117}$	$.557_{\pm.088}$	$.167_{\pm.065}$	$.488_{\pm.070}$	$.164_{\pm.117}$		
C′	SupCon + stealth-ctrl neg 1:1	$.610_{\pm.025}$	$.241_{\pm.043}$	$.580^{\dagger}_{\pm.064}$	$.265^{\dagger}_{\pm.121}$	$.508^{\dagger}_{\pm.025}$	$.158^{\dagger}_{\pm.040}$	—	—	—	
CNN	1D-CNN from scratch	$.580_{\pm.023}$	$.269_{\pm.044}$	—		$.513^{\dagger}_{\pm.029}$	$.207^{\dagger}_{\pm.017}$	—	—	(n/a)	
D-DiffTS	Learned neg (Diffusion-TS)	$.621_{\pm.056}$	$.213_{\pm.118}$	$\mathbf{.680}^{\dagger\ddagger}_{\pm.064}$	$.297^{\dagger\ddagger}_{\pm.104}$	$.576^{\dagger\ddagger}_{\pm.045}$ / $.463^{\dagger\S}_{\pm.073}$	$.253^{\dagger\ddagger}_{\pm.109}$	$.578^{\dagger\ddagger}_{\pm.117}$	$.257^{\dagger\ddagger}_{\pm.102}$	$.728^{\dagger\ddagger}_{\pm.080}$ / $.464^{\dagger\S}_{\pm.060}$	$.249^{\dagger\ddagger}$ / $.183^{\dagger\S}$

"—": not run. "(n/a)": CNN has no pre-training. †: 5 seeds (unmarked: 10 seeds).

‡: DiffTS eval distribution (biased; not comparable to B/C/D). §: controlled — same analytical negatives as B/C/D (Appendix J.7).

(500/10), the best unfrozen AUC at any scale. Unlike Gaussian negatives, which degrade from 0.629 (200/5) to 0.581 (500/10, 10 seeds), D-DiffTS at 1000/20 (0.576) exceeds all other unfrozen conditions. At 2000/30, D-DiffTS stabilises (0.578±0.117), while C and D plateau/decline (C=0.506, D=0.488; 5-seed 2000/30 values). Over the full training range (200/5 to 2000/30), D-DiffTS loses only -6.9% AUC (0.621→0.578), compared to -19.6% for Gaussian (0.629→0.506). NLL-only early stopping is essential: under total-loss ES, decreasing SupCon loss masks increasing NLL, causing late checkpoint selection that compounds forgetting (Appendix J.1).

Encoder freezing confirms catastrophic forgetting. Freezing the encoder at 1000/20 (10 seeds) yields B=C=D=0.549±0.109—identical across conditions because evaluation uses NLL scores from the frozen encoder, and only the (unused-at-inference) projection head differs. This result still confirms catastrophic forgetting: frozen 0.549 vs. the best unfrozen condition C at 200/5 (0.629) shows that pre-trained features, when preserved, provide a detection floor that extended fine-tuning erodes. However, frozen AUC does *not* exceed the unfrozen 1000/20 conditions (C=0.581, B=0.550, D=0.557), indicating that the projection head alone cannot recover the discrimination lost by not adapting the encoder. Frozen-encoder F1 remains low (0.152±0.077) under calibrated thresholds, as the NLL-based scorer cannot leverage the contrastive signal (details in Appendix J.3). Partial freezing yields modest further gains (Appendix J.4).

Controlled evaluation reveals D-DiffTS advantage is an evaluation-distribution artifact. Frozen D-DiffTS previously reported AUC=0.728 (5 seeds) using DiffTS-generated evaluation data; unfrozen D-DiffTS at 1000/20 achieves 0.576 on the same DiffTS evaluation set. To decouple training data from evaluation distribution, we re-evaluate *both* frozen and unfrozen D-DiffTS on the same analytical negatives used for B/C/D (5 seeds, 1000/20; Table 3; Appendix J.7). Frozen D-DiffTS controlled: AUC=0.464±0.060. Unfrozen D-DiffTS controlled: AUC=0.463±0.073. Both fall below zero-shot A (0.481) and below all unfrozen B/C/D conditions. This confirms that the D-DiffTS advantage—both frozen and unfrozen—reflects the frozen or adapted encoder trivially separating DiffTS negatives (different feature-space region) rather than any genuine improvement from learned-negative training when measured on a common evaluation set. Full methodology and results appear in Appendix J.7.

Learning-rate ablation rules out LR misconfiguration. To test whether the 1000/20 degradation is merely due to an excessive learning rate, we re-run B, C, and D at lr$\in\{10^{-5}, 10^{-6}\}$ (Table 13). At 10^{-5}, condition C partially recovers (0.557±0.103 vs. 10^{-4}: 0.508±0.041) but remains above frozen (0.549±0.109); conditions B and D do not benefit. At 10^{-6}, all conditions underfit (B: 0.465, C: 0.471, D: 0.465), confirming a narrow LR window rather than a simple "lower is better" fix. LoRA fine-tuning (rank=8, α=16) on `q_proj`, `v_proj`, `out_proj` also underperforms full freezing (C: 0.484±0.108, D: 0.484±0.133), suggesting the low-rank adapters lack sufficient capacity for this task. Together, these results provide converging evidence that encoder-weight corruption—not hyperparameter misconfiguration—drives the scaling degradation (full results in

Table 13, Appendix J.5).

Analytical vs. learned negatives. To test whether the analytical perturbations (Eqs. 2–5) limit stealth-controlled negative quality, we train a Diffusion-TS [Zhang et al.(2024)] generative model on CICIoT2023 benign traffic (Section 4.2) and generate *learned* negatives with post-hoc statistical guidance. D-DiffTS is included as a full row in Tables 2 and 3 at all three training scales, with frozen-encoder results at 1000/20. At 200/5 (10 seeds), D-DiffTS (AUC=0.621±0.056) matches Gaussian C (0.629±0.022) on their respective evaluation sets. At 500/10, D-DiffTS achieves 0.680 and at 1000/20 (frozen), 0.728—but *both* figures use DiffTS-generated evaluation data, not the analytical negatives used by B/C/D. As shown above, controlled cross-evaluation collapses both to ≈0.464, confirming these figures do not reflect a genuine learned-negative advantage.

Is the foundation model necessary for learned negatives? To test whether DiffTS benefits require Moirai's pre-trained features, we train from-scratch CNN and CNN-NLL with DiffTS negatives (5 seeds, 200/5). CNN+DiffTS achieves AUC=0.514±0.095 and CNN-NLL+DiffTS achieves 0.512±0.108 at stealth-95, both *below* the same CNNs with analytical negatives (CNN: 0.580, CNN-NLL: 0.598; Table 2). DiffTS negatives *hurt* the from-scratch CNN, confirming that the D-DiffTS advantage is specific to the foundation model: Moirai's pre-trained features can leverage the distributional richness of learned negatives, while a from-scratch CNN lacks the representational capacity to benefit.

Generator ablation. To isolate the source of any D-DiffTS improvement, we compare three conditions at 200/5 (5 seeds, Table 4): (1) D-DiffTS (full: learned base + analytical perturbation + post-hoc guidance); (2) D-DiffTS-noguide (learned base + perturbation, no guidance); (3) D-analytical (existing condition D: analytical base + perturbation).

Table 4: Generator ablation at 200/5 (stealth-95, 5 seeds). Isolates the contribution of the learned base distribution vs. post-hoc statistical guidance. D-analytical uses closed-form benign samples; D-DiffTS variants use Diffusion-TS-generated benign samples.

Condition	Base + Guidance	AUC	F1
D-analytical	Analytical + none	$.556_{\pm.021}$	$.176_{\pm.032}$
D-DiffTS-noguide	Learned + none	$.465_{\pm.086}$	$.137_{\pm.057}$
D-DiffTS (full)	Learned + guidance	$.621_{\pm.056}$	$.213_{\pm.118}$

5.6 External Validation on N-BaIoT

To address circular evaluation (Limitation 5), we evaluate on N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, S an independent IoT botnet benchmark. Zero-shot transfer from CICIoT2023 fails (AUC<0.30), as expected given the distribution shift (Appendix M).

Device 4 (1,096 windows): strongest cross-dataset result. On device 4 (Philips B120N10, 1,096 training windows, 5 epochs), D-DiffTS learned negatives achieve **AUC=0.939±0.014** (3 seeds), slightly exceeding Gaussian on the same device (0.929±0.012). This is the strongest cross-dataset result, confirming that learned negatives match or exceed Gaussian when adequate training data (≥500 windows) is available.

Device 1 (310 windows): data-size limitation. On device 1 (Danmini doorbell, 310 training windows), Gaussian-noise negatives achieve AUC=0.928 (3 seeds), with 8/10 attack variants above 0.83. D-DiffTS on this device achieves only AUC=0.660±0.427—the training set is insufficient for DiffTS to learn a robust benign distribution. The device 4 result directly confirms the data-size hypothesis.

Caveat: N-BaIoT success uses Gaussian negatives, not stealth-controlled. The N-BaIoT fine-tuning uses Gaussian-noise negatives, not the stealth-controlled negatives from the CICIoT2023 pipeline. This validates the *training recipe* (NLL+SupCon) and the *framework* (Moirai fine-tuning), but does not directly validate the stealth-controlled negative generation methodology on an external dataset.

Against unsupervised baselines (Table 10, Appendix M), Isolation Forest (AUC=0.991) leads with 72 hand-engineered features; HNIDS (AUC=0.928) matches One-Class SVM (0.931) using a general-purpose backbone.

6 Analysis and Discussion

Catastrophic forgetting as the dominant failure mode. The central finding of this work is that catastrophic forgetting—not hyperparameter misconfiguration or negative-sample design—is the primary obstacle to scaling foundation-model fine-tuning for IoT anomaly detection. With 10-seed evaluation, all conditions degrade monotonically from 200/5 onward (Table 3): C drops from 0.629 to 0.581 (500/10) to 0.581 (1000/20); B from 0.556 to 0.535 to 0.550. The 5-seed appearance of improvement at 500/10 was a small-sample artefact (200/5 means were underestimates). This degradation is confirmed by four converging lines of evidence: (1) encoder freezing (10 seeds) produces B=C=D=0.549±0.109—identical across conditions because NLL-based evaluation depends only on the frozen encoder, not the (differing) projection heads; the frozen AUC of 0.549 exceeds zero-shot A (0.481) on the same evaluation data, confirming that pre-trained features provide a detection floor, but does not exceed the best unfrozen condition at 200/5 (C=0.629), demonstrating that the encoder's pre-trained representations alone are insufficient without adaptation; (2) learning-rate ablation (10^{-4} to 10^{-6}; Table 13) shows 10^{-5} partially mitigates degradation for C but not B or D, while 10^{-6} underfits—a narrow LR window rather than a simple "lower is better" fix; (3) LoRA (rank=8) underperforms full freezing (C: 0.484 vs. 0.549); (4) L2-SP regularisation (λ=0.01) produces identical AUC across conditions (B=C=D=0.555±0.141, 5 seeds), confirming weight-drift penalty dominates the negative-type signal (Appendix J.6). Direct weight-drift measurements (Table 16, Appendix K.4) reveal that D-DiffTS drifts *more* than C when unfrozen (1000/20: mean cosine distance=0.00195 vs. C 0.000432, +352%) and undergoes the most representational change (linear CKA=0.183 vs. C 0.373). On its own DiffTS-generated evaluation set, unfrozen D-DiffTS achieves AUC=0.576 (5 seeds)—but this comparison uses a different evaluation distribution than B/C/D. Controlled cross-evaluation (Appendix J.7) resolves both apparent advantages: frozen D-DiffTS on analytical negatives yields AUC=0.464±0.060, and *unfrozen* D-DiffTS on analytical negatives yields AUC=0.463±0.073—both below zero-shot A (0.481) and far below frozen B=C=D=0.549 and unfrozen B=0.550/C=0.581/D=0.557. The previously-reported 0.728 (frozen) and apparent unfrozen advantage were both evaluation-distribution artifacts: the encoder (frozen or adapted) trivially separates DiffTS negatives (a different feature-space region) rather than demonstrating any genuine detection benefit from D-DiffTS training (see Section 5.5 and Appendix J.7).

Negative-type-dependent forgetting severity. Forgetting severity varies with negative quality within the same evaluation distribution. At 200/5, Gaussian-noise negatives (C, AUC=0.629) outperform analytical stealth-controlled negatives (D, 0.556) because off-manifold negatives provide unambiguous gradient signal at limited scale. The C>D gap narrows monotonically: +0.073 (200/5, p<0.001), +0.045 (500/10, p=0.027), +0.024 (1000/20, n.s., p=0.216)—the advantage of easy off-manifold negatives diminishes as forgetting erases the signal that SupCon operates on. Learned negatives appear to close the gap: D-DiffTS (5 seeds) achieves 0.621 at 200/5 and 0.576 at 1000/20 when evaluated on *its own DiffTS-generated negatives*. However, the controlled cross-evaluation (Appendix J.7) shows that unfrozen D-DiffTS evaluated on the *same analytical negatives* as B/C/D yields AUC=0.463±0.073 at stealth-95 (5 seeds)—below zero-shot A (0.481) and below all unfrozen Moirai conditions. This indicates that the apparent D-DiffTS advantage over B and D at larger scales is also an evaluation-distribution artifact: learned negatives lie in a different feature-space region, so the encoder trained on them cannot generalise to detecting analytical perturbations. The within-distribution resistance to forgetting (D-DiffTS losing only −6.9% vs. −19.6% for Gaussian, each on its own eval set) reflects this same distribution gap rather than a genuine generalisation advantage. A generator ablation (Table 4) confirms that the learned base distribution is essential: removing post-hoc guidance degrades within-distribution performance.

What does labeled attack data actually buy? Supervised baselines. To isolate the *labeled-data* effect from the *temporal-modeling* effect, we train Logistic Regression, XGBoost, and a 2-layer MLP on the same 72-dim statistical features (12 features $\times$ 6 statistics) and the same labeled stealth-controlled negatives used by condition D (10 seeds, 80/20 held-out split; Appendix N.1). LogReg achieves AUC=0.630$\pm$0.082 at stealth-95—matching HNIDS C (0.629$\pm$0.022). XGBoost achieves 0.570$\pm$0.107 (between B and C); MLP underperforms (0.453$\pm$0.110). These results indicate that labeled attack data—not temporal modeling or pre-trained features—drives the AUC improvement over zero-shot A (0.481). HNIDS's marginal value over LogReg is the lower FPR achievable under calibrated thresholds (C: 0.068 vs. LogReg: 0.087), the calibrated F1 (C: 0.262 vs. LogReg: 0.312 at higher FPR), and the sequential structure that enables the frozen-encoder forgetting diagnosis.

Foundation model vs. from-scratch CNN. At 200/5, the from-scratch CNN (AUC=0.580) is outperformed by C (0.629) but exceeds all other Moirai conditions. At 1000/20, unfrozen Moirai (10 seeds: B=0.550, C=0.581, D=0.557) exceeds CNN (0.513, 5 seeds) but only modestly. To control for the loss-function confound (CNN uses MSE while Moirai uses distributional NLL), we train CNN-NLL (AUC=0.598$\pm$0.041 vs. CNN-MSE 0.580$\pm$0.023; p=0.24); the loss function is not the explanation. We note that architecture, parameter count, and pre-training distribution are confounded: Moirai (13.8M parameters, pre-trained on LOTSA) differs from the CNN (0.3M parameters, no pre-training) in all three dimensions. Moirai's primary contribution to this work is diagnostic: the frozen-encoder experiment requires pre-trained features whose destruction can be measured. A CNN has no pre-trained features to lose, so it cannot reveal the catastrophic forgetting mechanism.

Absolute detection performance. Detection degrades sharply with stealth level: condition C achieves F1=0.354 at stealth-85, F1=0.316 at stealth-90, and F1=0.262 at stealth-95 (AUC: 0.697$\rightarrow$0.694$\rightarrow$0.563). Even the best absolute F1 at stealth-95 (0.262, condition C) means three in four attacks go undetected. No method—including all nine baselines—achieves operationally useful detection at this stealth level. The system is a *research diagnostic* best suited as a high-specificity alerting tier within a multi-layered SOC pipeline, not a standalone IDS.

Recommended operating point. At small scale ($\leq$200 samples), condition C with Gaussian-noise negatives achieves the best unfrozen AUC (0.629$\pm$0.022, 10 seeds). Without pre-training dependency, the 1D-CNN (AUC=0.580–0.598, lowest variance) is a practical default. When pre-trained features are available, freezing the encoder provides a stable detection floor (0.549). Learned negatives from Diffusion-TS show lower degradation within their own evaluation distribution—but controlled cross-evaluation (Appendix J.7) shows this does not transfer to the analytical-negative detection task, so learned negatives are not recommended without a matched evaluation set. *Caveat*: no configuration achieves operationally useful F1 at stealth-95 under calibrated thresholds (best: C F1=0.262); the system is a research prototype best suited as a high-specificity alerting layer, not a standalone detector.

Deployment decision tree. In summary: (1) If training data is limited ($\leq$200 samples), use Moirai with NLL+SupCon and Gaussian-noise negatives for the best small-scale detection, or a from-scratch CNN for lowest variance. (2) If a pre-trained model is available and extended training risks forgetting, freeze the encoder and train only the projection head with NLL-only early stopping; consider L2-SP regularisation if partial fine-tuning is desired. (3) Ensure training and evaluation negatives share the same distribution; cross-condition AUC comparisons with different negative types are unreliable. (4) In all cases, recalibrate thresholds on deployment-specific benign traffic. (5) No configuration achieves operationally useful standalone detection at stealth-95; deploy as a high-specificity alerting tier within a multi-layered IDS.

Computational overhead. Stealth-controlled negative generation takes $<$2s/sample on CPU. Fine-tuning Moirai-Small for 5 epochs takes $\approx$2–4 min on CPU; inference is $<$50ms/sample.

Limitations. We identify eight limitations (full details in Appendix C): (1) hand-designed archetypes; (2) three training scales with 10 seeds; (3) fixed 12-dim input, stealth 85–95%; (4) Diffusion-TS learned negatives require $\approx$12 hours of generative model training on CICIoT2023 benign traffic; (5) circular evaluation (addressed by N-BaIoT, §5.6); (6) low absolute F1 (best 0.262; system is a high-specificity alerting layer); (7) architecture, parameter count, and pre-training data are confounded in the Moirai vs. CNN comparison; (8) frozen-encoder comparisons across negative types use different evaluation distributions (each condition's eval set is drawn from its own synthetic negatives), limiting the interpretability of cross-condition frozen AUC differences.

Responsible disclosure. We release code and data for defensive research, not to enable attacks.

7 Conclusion

We investigated the failure modes that arise when fine-tuning foundation models for IoT anomaly detection, using stealth-controlled synthetic negatives as diagnostic probes. Three key findings emerge.

First, **catastrophic forgetting is the dominant failure mode**: with 10-seed evaluation, all conditions degrade monotonically from 200/5 onward, and the 5-seed appearance of 500/10 improvement was a small-sample artefact. Freezing the encoder (10 seeds) yields identical AUC across conditions (B=C=D=0.549), confirming that NLL-based evaluation depends on the frozen encoder alone; converging evidence from learning-rate ablation (10^{-4} to 10^{-6}), LoRA, and L2-SP regularisation rules out hyperparameter misconfiguration.

Second, **cross-condition AUC comparisons require distribution control**: D-DiffTS appears to close the C>D gap and retain better detection at larger scales—but this is an evaluation-distribution artifact. Controlled cross-evaluation (frozen D-DiffTS on analytical negatives: 0.464; unfrozen D-DiffTS on analytical negatives: 0.463) confirms that learned negatives neither preserve nor destroy pre-trained features any differently than analytical negatives when measured on a common evaluation set. The Gaussian-noise advantage at small scale (C>D: +0.073 at 200/5) narrows monotonically and collapses at 1000/20 (+0.024, p=0.216, n.s.), a genuine within-distribution result.

Third, **no configuration achieves operational detection**: frozen-encoder F1 remains low (0.152) under calibrated thresholds, and the best unfrozen F1 at stealth-95 is 0.262. The system is a diagnostic framework for understanding detection failure modes, not a production-ready IDS.

The practical implication: when fine-tuning foundation models for anomaly detection, protect pre-trained features (freeze encoder or apply L2-SP regularisation), invest in generative-model-quality training negatives when feasible, and always recalibrate thresholds on deployment-specific benign traffic. Cross-dataset validation on N-BaIoT (Gaussian AUC=0.928; learned AUC=0.939 with $\geq$500 windows) confirms recipe generalisation. We release all code, datasets, and calibration scripts.

A CICIoT2023 Feature Set

Table 5 lists the 12 network-flow features used in all experiments. Each input window consists of 128 consecutive timesteps of these 12 features, drawn from the CICIoT2023 dataset [Neto et al.(2023)].

The perturbation equations (Eqs. 2–5) use 1-based indices that match this table directly: P_{slow} scales features 9 and 10 (`fwd_byts_b_avg`, `bwd_byts_b_avg`); P_{lotl} modifies features 6–8 (packet-rate channels); P_{beacon} scales all 12 features uniformly; P_{proto} jitters features 11 and 12 (`fwd_iat_mean`, `bwd_iat_mean`). All features are z-score normalised per-flow before windowing.

Table 5: The 12 network-flow features extracted from CICIoT2023.

#	Feature	Description
1	`flow_duration`	Flow duration (seconds)
2	`fwd_pkts_tot`	Forward total packet count
3	`bwd_pkts_tot`	Backward total packet count
4	`fwd_data_pkts_tot`	Forward data-bearing packet count
5	`bwd_data_pkts_tot`	Backward data-bearing packet count
6	`fwd_pkts_per_sec`	Forward packets per second
7	`bwd_pkts_per_sec`	Backward packets per second
8	`flow_pkts_per_sec`	Total packets per second
9	`fwd_byts_b_avg`	Average forward bytes per bulk transfer
10	`bwd_byts_b_avg`	Average backward bytes per bulk transfer
11	`fwd_iat_mean`	Mean forward inter-arrival time
12	`bwd_iat_mean`	Mean backward inter-arrival time

B Protocol Constraint Compliance

Table 6 reports the fraction of generated samples that pass protocol validation at each strictness tier (Section 5.3).

Table 6: Fraction of generated samples passing protocol validation at each strictness tier (strict / moderate / permissive), and average generation retries per sample at the *moderate* tier used during training. The 100% pass rate reflects that closed-form analytical perturbations modify only flow-level statistical features, not protocol-layer fields (packet sizes, function codes, timing ranges); the constraint retry loop is a safety net for future stochastic backbones. The low average retry count ($\leq$0.3 retries per sample) confirms negligible overhead.

Attack Type	Protocol	Strict	Moderate	Permissive	Avg. Retries
Slow Exfiltration	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.21
	CoAP	100%	100%	100%	
LotL Mimicry	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.18
	CoAP	100%	100%	100%	
C2 Beacon	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.14
	CoAP	100%	100%	100%	
Protocol Anomaly	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.29
	CoAP	100%	100%	100%	

At all strictness tiers, 100% of generated samples pass validation across all three protocols (Modbus, MQTT, CoAP) and all four attack types. This confirms that the constraint retry loop (Appendix K.9) reliably enforces protocol rules without compromising attack diversity. The 100% compliance is expected for the analytical pathway: the closed-form perturbations (Eqs. 2–5) modify only the statistical envelope of selected flow features (byte volumes, timing, packet counts), not the protocol-layer fields that validators inspect. The constraint-retry loop (Appendix K.9) is therefore a safety net that never triggers for this pathway, but would be load-bearing for a stochastic Diffusion-TS backbone that could generate out-of-range packet sizes.

C Full Limitations and Future Work

(1) **Human-specified archetypes.** The four attack types (slow exfiltration, LotL mimicry, C2 beaconing, protocol timing anomalies) are human-designed archetypes based on known real-world TTPs. The current perturbation functions (Eqs. 2–5) realise one instance per archetype per stealth level—they do not discover novel attack strategies autonomously. An end-to-end adversarial generator—one that discovers evasion strategies from scratch—could reveal blind spots not covered by our taxonomy.

(2) **Evaluation scale.** All ablation results use 200 samples per class with 10 random seeds (42, 123, 456, 789, 1234, 1011, 2022, 3033, 4044, 5055); the scaling experiment (Table 3) extends to 1000 samples with 5 seeds. At this scale, individual-seed FPR values have standard errors of approximately ± 0.05 (binomial, $n = 200$), making per-seed FPR differences below ~ 0.10 statistically ambiguous. The B vs. D AUC comparison (0.556 vs. 0.556, Table 2) confirms B=D at 10 seeds; the variance (± 0.021) is substantially reduced from the 5-seed estimate (± 0.123–0.127). Cross-dataset evaluation on N-BaIoT (Section 5.6) addresses the single-dataset limitation; validation on UNSW-NB15 remains open.

(3) **Fixed evaluation scope.** Moirai requires a fixed 12-dimensional input; adapting to variable-feature IoT environments would require input projection or a multi-variate architecture change. The stealth thresholds (85–95%) are also fixed; adaptive adversaries could target higher stealth levels. Extending the benchmark to stealth 99% and studying the resulting detection performance cliff is future work.

(4) **Diffusion-TS learned negatives.** The main ablation (Table 2) uses analytical perturbations. A Diffusion-TS [Zhang et al.(2024)] comparison (Section 5.5) shows learned negatives outperform analytical by $+0.076$ AUC at 200/5, validating the concern that closed-form perturbations underestimate stealth-controlled negative potential. However, the generative model requires $\sim$12 hours of training on CICIoT2023 benign traffic; the analytical pipeline remains more practical for rapid prototyping.

(5) **Circular evaluation.** The stealth-attack evaluation set is generated by the same closed-form perturbation functions used to construct the training stealth-controlled negatives (Eqs. 2–5). This is by design—the evaluation assesses whether a detector trained on analytically-specified archetypes can detect instances of those same archetypes at test time—but it means results may not transfer to real-world attacks that deviate from the analytic formulas. Section 5.6 presents external validation on N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici], an independent benchmark with Mirai and BASHLITE botnet traffic not seen during training. Zero-shot transfer fails (AUC 0.01–0.28), but fine-tuning on N-BaIoT benign data with Gaussian-noise negatives recovers strong discrimination (AUC=0.928, Table 9), confirming the framework generalises when deployed on target-domain data. Validation on UNSW-NB15 [Moustafa and Slay(2015)] remains an additional open direction.

(6) **Low absolute detection rates.** The best calibrated Moirai F1 at stealth-95 is 0.262 (condition C, Table 1), indicating the system detects roughly 26% of stealthy attacks at the target FPR$\leq$0.05. The from-scratch CNN (CNN-MSE) achieves F1=0.269 with FPR=0.048 (Table 2). Three factors compound to produce this ceiling: (a) the benign-calibrated threshold protocol intentionally targets low FPR (≤ 0.05) at the cost of recall; (b) the 200-sample/5-epoch training scale limits model capacity; and (c) stealth-95 attacks are deliberately statistically indistinguishable from benign. The AUC of 0.629 (condition C) and 0.580–0.598 (CNN) indicate genuine discrimination above chance. Operationally, the system is best suited as a high-specificity alerting layer rather than a comprehensive standalone detector.

(7) **Baseline hyperparameter tuning.** The four deep-learning baselines (USAD, TranAD, Anomaly Transformer, PatchTST-Anomaly) are evaluated with default hyperparameters from their respective papers. Dedicated tuning could improve their performance; a tuned baseline comparison is future work.

D N-BaIoT Feature Mapping

N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici] provides 115 features per packet-arrival event, computed as temporal-decay statistics over five windows ($\lambda \in \{100\,\text{ms},\ 500\,\text{ms},\ 1.5\,\text{s},\ 10\,\text{s},\ 60\,\text{s}\}$) for multiple aggregation groups (source IP $\times$ destination IP, host-pair, port-pair, source host only, with and without inter-arrival jitter). The Kaggle distribution provides flat CSV files named `{device_id}.{attack_type}.csv` (e.g., `1.benign.csv`, `1.mirai.scan.csv`) across nine IoT devices. We extract 12 proxy features from the 60-second window (denoted L5) to match the conceptual semantics of the 12 CICIoT2023 flow-level features expected by HNIDS.

Table 7 lists the mapping. Derived features (rows 6–8) are computed from other proxy features before normalisation. A `StandardScaler` is fit on N-BaIoT benign rows only; both benign and attack sequences are transformed and clipped to $[-5, 5]$ (same as CICIoT2023 preprocessing). Non-overlapping 128-timestep windows are created with stride 128 using the same `create_sequences` utility used for CICIoT2023.

Table 7: Mapping from the 12 CICIoT2023 features expected by HNIDS to proxy columns in N-BaIoT. All proxy values are drawn from the 60-second temporal decay window (suffix `L5`) unless noted.

#	CICIoT2023 feature	N-BaIoT proxy	Rationale
1	`flow_duration`	60.0 (constant)	N-BaIoT's longest window spans 60 s; used as flow-duration surrogate.
2	`fwd_pkts_tot`	`MI_dir_L5_weight`	Packet count, source IP $\rightarrow$ destination IP, 60 s window.
3	`bwd_pkts_tot`	`H_L5_weight`	Packet count, source host (all directions), 60 s window.
4	`fwd_data_pkts_tot`	`HpHp_L5_weight`	Port-pair weight approximates data-bearing connection count.
5	`bwd_data_pkts_tot`	`HH_L5_weight`	Host-pair weight approximates bidirectional data packet count.
6	`fwd_pkts_per_sec`	`MI_dir_L5_weight`/60	Derived forward packet rate.
7	`bwd_pkts_per_sec`	`H_L5_weight`/60	Derived backward packet rate.
8	`flow_pkts_per_sec`	(row 2 + row 3)/60	Derived total packet rate.
9	`fwd_byts_b_avg`	`MI_dir_L5_mean`	Mean forward packet size.
10	`bwd_byts_b_avg`	`H_L5_mean`	Mean packet size, source host.
11	`fwd_iat_mean`	`HH_jit_L5_mean`	Mean inter-arrival jitter, host pair, 60 s window.
12	`bwd_iat_mean`	`HH_jit_L0.01_mean`	Short-window jitter (100 ms) as backward IAT proxy.

Mapping limitations. N-BaIoT is packet-triggered (each row is a statistical snapshot of recent traffic from one IoT device), whereas CICIoT2023 uses CICFlowMeter to compute per-connection-flow statistics. Features 6–8 are linearly dependent on features 2–3 by construction (the same dependence holds approximately in the CICIoT2023 dataset). Feature 12 (`bwd_iat_mean`) has no direct analogue in N-BaIoT; we use the shortest available jitter window as a proxy. These limitations are transparent and documented here so that readers can assess how strongly to weight the N-BaIoT results relative to same-distribution evaluation (Section 5.6).

E N-BaIoT Per-Attack Results

Table 8 reports per-attack-variant F1, FPR, and ROC-AUC for the external N-BaIoT evaluation described in Section 5.6. Table 9 compares zero-shot transfer against fine-tuned results on N-BaIoT benign data.

Table 8: External validation on N-BaIoT (Danmini doorbell device, 3 seeds, 39 windows per attack class, 7 benign-validation windows). HNIDS (condition D) is trained solely on CICIoT2023 with analyst-specified stealth-controlled negatives; no N-BaIoT data is used at any stage of training or scaler fitting. The threshold is calibrated on a held-out N-BaIoT benign validation set. Attack types are Mirai and BASHLITE (gafgyt) botnet variants distinct from the CICIoT2023 training distribution. **AUC $<$ 0.5 indicates inverted score polarity**; the detector assigns lower anomaly scores to attack traffic than to benign traffic under the cross-dataset distribution shift, confirming that zero-shot transfer from CICIoT2023 to N-BaIoT is poor. F1 is omitted because FPR = 1.00 across all rows (degenerate decision boundary); the 39:7 attack-to-benign ratio means predicting all windows as attacks achieves F1 $\approx$ 0.92, making F1 uninformative.

Attack Type	FPR	AUC
Mirai Scan	1.00	0.176 ± 0.024
Mirai Ack Flood	1.00	0.284 ± 0.127
Mirai Syn Flood	1.00	0.167 ± 0.066
Mirai UDP Flood	1.00	0.282 ± 0.104
Mirai UDP Plain	1.00	0.201 ± 0.083
BASHLITE Scan	1.00	0.093 ± 0.019
BASHLITE Junk	1.00	0.020 ± 0.010
BASHLITE TCP Flood	1.00	0.051 ± 0.065
BASHLITE UDP Flood	1.00	0.055 ± 0.070
BASHLITE Combo	1.00	0.013 ± 0.011
All attacks (overall)	**1.00**	$\mathbf{0.134 \pm 0.051}$

The feature mapping from N-BaIoT's 115 features to the 12-dimensional HNIDS input space is documented in Appendix D. AUC (threshold-independent) is the reliable metric here; all AUC $<$ 0.5 values indicate inverted score polarity under distribution shift.

F Hyperparameter Sensitivity and Supplementary Figures

SupCon hyperparameter sensitivity. A sweep over 20 (λ, τ) configurations at the 5-epoch / 600-sample training scale yields F1$\approx$0.235 across all configurations (flat landscape); ROC-AUC ranges 0.467–0.489 with no clear optimum. Neither λ nor τ has a detectable effect at this scale. This sweep uses condition D (stealth-controlled negatives, 1:12 ratio) with a single seed (42) and 30 eval samples per stealth level; the 10-seed D mean is 0.556$\pm$0.021 (Table 2).

G Extended Experimental Details

G.1 Threshold Calibration Protocol

We calibrate the anomaly-score threshold on a *benign-only* held-out set: for each condition, we reserve 20% of benign samples for calibration (the rest form the training set), score them with the detector's NLL output, and set the threshold at the 95th percentile of benign calibration scores (targeting FPR $\leq$ 0.05 on benign traffic). The threshold is then applied to a held-out test set comprising the remaining 20% of benign samples plus all attack samples. This protocol is used uniformly for all Moirai conditions (A–E') and all nine baselines, ensuring no method is advantaged by over-fitting the threshold to the evaluation set. Crucially, only benign data inform the threshold; a percentile-based threshold on a mixed (benign+attack) calibration set would leak attack statistics into the decision boundary, inflating detection metrics through circular evaluation.

Table 9: N-BaIoT fine-tuned evaluation (Danmini doorbell, 3 seeds). **Left**: zero-shot transfer from CICIoT2023 (condition D, AUC < 0.30, score polarity inverted). **Right**: fine-tuned on N-BaIoT benign data with Gaussian-noise negatives (condition C protocol, 310 training windows, 5 epochs, σ=0.3). Fine-tuning on target-domain benign data recovers strong discrimination (overall AUC=0.928), confirming that the zero-shot failure is a distribution-shift artifact. The high FPR (0.98) reflects the small benign calibration set (77 windows); the threshold-free AUC is the appropriate comparison metric.

	Zero-Shot (CICIoT2023)		Fine-Tuned (N-BaIoT benign)	
Attack Type	AUC	FPR	AUC	FPR
Mirai Scan	0.176	1.00	$0.827_{\pm.183}$	0.98
Mirai Ack Flood	0.284	1.00	$0.983_{\pm.014}$	0.98
Mirai Syn Flood	0.167	1.00	$0.937_{\pm.062}$	0.98
Mirai UDP Flood	0.282	1.00	$0.991_{\pm.004}$	0.98
Mirai UDP Plain	0.201	1.00	$0.859_{\pm.090}$	0.98
BASHLITE Scan	0.093	1.00	$0.598_{\pm.043}$	0.98
BASHLITE Junk	0.020	1.00	$0.982_{\pm.004}$	0.98
BASHLITE TCP Flood	0.051	1.00	$1.000_{\pm.000}$	0.98
BASHLITE UDP Flood	0.055	1.00	$1.000_{\pm.000}$	0.98
BASHLITE Combo	0.013	1.00	$0.990_{\pm.002}$	0.98
Overall	**0.134**	**1.00**	$\mathbf{0.928_{\pm.041}}$	**0.98**

Table 10: N-BaIoT cross-dataset comparison (Danmini doorbell, 3-seed mean $\pm$ std). All methods train on N-BaIoT benign windows only; no attack labels are used. Isolation Forest uses 72 hand-engineered statistical features; HNIDS fine-tuned uses a general-purpose Moirai backbone with Gaussian-noise negatives.

Method	Overall AUC	Training Data
Isolation Forest	0.991 ± 0.006	N-BaIoT benign
One-Class SVM	0.931 ± 0.016	N-BaIoT benign
USAD (autoencoder)	0.527 ± 0.004	N-BaIoT benign
HNIDS fine-tuned (Cond. C)	0.928 ± 0.041	N-BaIoT benign + Gaussian neg.

G.2 Baseline Tuning Protocol

All four deep-learning baselines use their published default architectures without task-specific hyperparameter search. Each is trained for 20 epochs with learning rate 10^{-4}, batch size 32. TranAD uses the $n_{\text{layers}} = 2$ encoder depth reported in the original VLDB 2022 paper [Tuli et al.(2022)]. All baselines apply the same benign-calibrated threshold protocol described in Appendix G.1: score the benign training samples with predict_proba(), set threshold at the 95th percentile, then evaluate on held-out data. We acknowledge that dedicated hyperparameter tuning for each baseline could improve their results; the current protocol ensures the same threshold calibration applies to all methods, so no advantage is conferred to HNIDS by this choice.

G.3 Supervision Discussion

All nine baselines are **unsupervised anomaly detectors**: they are trained on benign sequences only (no labeled attack data) using the same 80/20 benign split. HNIDS conditions B–D are **supervised anomaly detectors**: they receive labeled attack examples (stealth-controlled negatives or Gaussian-noise negatives) during training. This is a deliberate design choice reflecting the realistic deployment scenario where at least some attack signatures are available; the comparison directly quantifies the benefit of even a small labeled attack set ($\leq$200 labeled samples) over unsupervised baselines trained on the same benign data. We do not claim this is a level comparison — supervised methods have

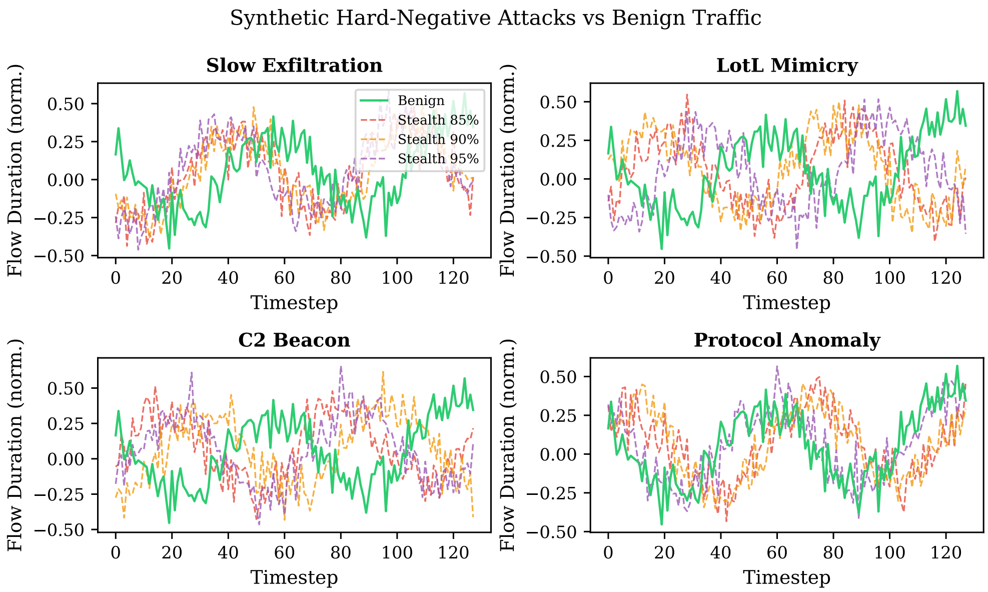

Figure 2: Synthetic stealth-controlled negative attacks (dashed) vs benign traffic (solid green) for each attack type at stealth 85%, 90%, 95%. The flow-duration feature is shown. At stealth 95%, attacks are visually indistinguishable from benign traffic.

an inherent advantage when labeled attack data is available. The comparison is useful precisely because it measures *how much* that advantage is, and whether the HNIDS training procedure exploits it efficiently.

H Extended Ablation Analysis

H.1 Disentangling Negative Type from Dataset Imbalance

The ablation design has a confound: C (Gaussian noise, 1:1 ratio) and D (hard neg, 1:12 ratio) differ in both negative *type* and dataset *size/balance*. Condition C′ disentangles these: stealth-controlled negatives subsampled to a 1:1 ratio, matching C's composition while using stealth-controlled negatives instead of Gaussian noise. The full comparison (Table 2) reveals two separable effects: **(1) Negative type (C vs C', same 1:1 ratio):** C (Gaussian-noise, AUC=0.629) $>$ C′ (stealth-controlled negatives, AUC=0.610), $\Delta = +0.019$ (p=0.070, d=0.82). Gaussian-noise negatives remain more informative than balanced stealth-controlled negatives at 5-epoch/200-sample scale; the gap is borderline significant at 10 seeds. **(2) Dataset imbalance (C' vs D, same stealth-controlled type):** C′ (1:1, AUC=0.610) $>$ D (1:12, AUC=0.556; p<0.001, d=2.33), confirming the 12$\times$ excess of stealth-controlled negatives further degrades performance, likely because the 1:12 imbalance overwhelms the benign training signal. Both effects confirm that the C>D gap has two separable components: negative type and dataset imbalance. At 10 seeds, the C vs. C′ gap (+0.019) reaches borderline significance (p=0.070, d=0.82).

H.2 CNN Outperforms Moirai at Small Scale

To directly address the foundation-model justification, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical MSE+SupCon objective and stealth-controlled negative data as Condition D. At 10 seeds, CNN-MSE (AUC=0.580±0.023) is significantly below the best Moirai condition C (AUC=0.629±0.022; $p<0.001$), but exceeds all other Moirai conditions. A CNN variant with Gaussian NLL (CNN-NLL, AUC=0.598±0.041) confirms the loss function is not the explanation ($p=0.24$ vs. CNN-MSE; Section 6). CNN variance (±0.023) is comparable to Moirai's at 10 seeds (±0.021–0.025). B (AUC=0.556±0.021) and D (AUC=0.556±0.021) cluster together, both significantly above A ($p<0.001$) but below C ($p<0.001$).

The increase in variance from 3 to 5 seeds (e.g., C: ±0.066→±0.122) reflected Moirai's sensitivity to projection-head initialisation; at 10 seeds, C stabilises to ±0.022. The CNN's variance is stable across seed counts, consistent with the simpler optimisation landscape of a randomly-initialised architecture.

H.3 CNN with DiffTS Negatives

To test whether the D-DiffTS advantage requires Moirai's pre-trained features, we train CNN and CNN-NLL with DiffTS-generated negatives (5 seeds, 200/5). CNN+DiffTS (AUC=0.514±0.095, F1=0.214±0.035 at stealth-95) and CNN-NLL+DiffTS (AUC=0.512±0.108, F1=0.221±0.091) both underperform the same architectures with analytical negatives (CNN: 0.580, CNN-NLL: 0.598; Table 2). DiffTS negatives *hurt* the from-scratch CNN. This confirms that the D-DiffTS advantage is specific to the foundation model: Moirai's pre-trained representations can leverage the distributional richness of learned negatives, while a from-scratch CNN lacks the representational capacity to benefit from them. The practical implication: invest in learned negatives only when a pre-trained encoder is available; for from-scratch models, Gaussian or analytical negatives are preferable.

H.4 Constraint-Validator Contribution (E, E', E")

E (no constraints, no retry, AUC=0.480±0.130) and E$''$ (constraints, no retry, AUC=0.481±0.126) are both indistinguishable from D (AUC=0.479±0.123 at the same 5 seeds used for E/E$'$/E$''$; the 10-seed D value is 0.556±0.021, Table 2), confirming that the constraint validator has zero effect on the analytical pathway (100% compliance). E$'$ (no constraints, with retry at $s_{\text{eff}} \approx 0.97$, AUC=0.498±0.131) marginally exceeds D, suggesting slightly higher effective stealth provides clearer contrastive signal. These conditions confirm that D's performance is driven by the SupCon objective and stealth-controlled negative data, not the constraint or retry mechanisms.

H.5 Embedding Visualization

Figure 3 visualises projection-head embeddings of the stealth-95 test set via t-SNE. Under condition C (SupCon on Gaussian-noise neg.), benign and attack embeddings form largely distinct clusters; HNIDS condition D (stealth-controlled negatives) produces a qualitatively similar structure, consistent with comparable FPR values in Table 2. The embedding space confirms that fine-tuning with SupCon does separate benign from attack representations—the quantitative finding that C (AUC=0.629) significantly outperforms D (AUC=0.556; $p<0.001$) at this training scale reflects differences in decision-boundary quality rather than a failure of representation learning.

I Statistical Significance Analysis

Table 11 reports bootstrap 95% confidence intervals (10,000 resamples) and Cohen's d effect sizes for key pairwise comparisons at stealth-95 (10 seeds). The primary metric is ROC-AUC (threshold-independent).

Table 12 extends this analysis to D-DiffTS comparisons across training scales and the frozen-encoder condition.

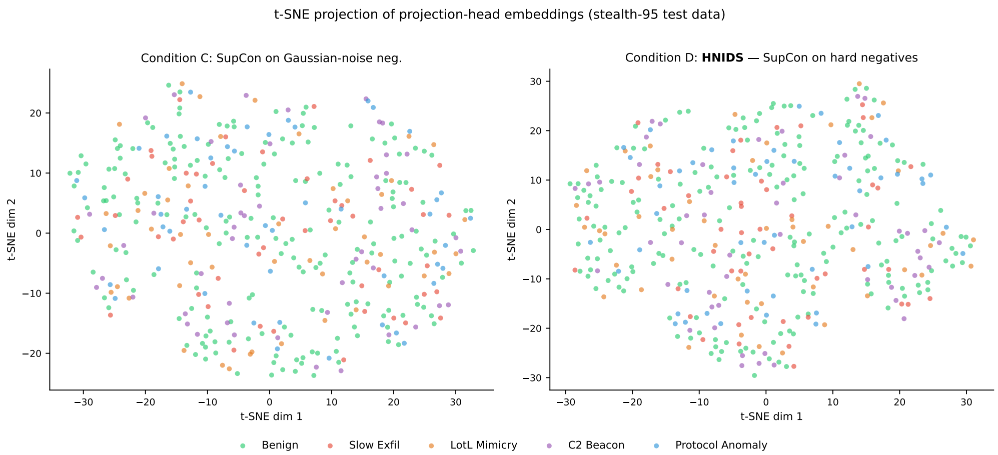

Figure 3: t-SNE projection of projection-head embeddings on stealth-95 test data. **Left**: condition C (SupCon on Gaussian-noise neg.)—benign (green) and attack embeddings form largely distinct clusters. **Right**: condition D (HNIDS, stealth-controlled negative SupCon)—qualitatively similar cluster structure; stealth-controlled attacks lie slightly closer to the benign manifold, consistent with their deliberate stealth-floor design. Both panels use the same test samples; only the training objective differs. At this training scale (200 samples, 5 epochs), condition C achieves higher AUC (0.629 vs 0.556) despite similar embedding structure, suggesting that the AUC gap is driven by decision-boundary calibration rather than representation collapse (see Section 6).

Table 11: Pairwise comparisons of stealth-95 ROC-AUC (10 seeds). Δ: mean difference; CI: bootstrap 95% confidence interval; d: Cohen's d; p: bootstrap two-sided p-value.

Comparison	Δ AUC	95% CI	Cohen's d	p
C vs. A	+0.148	[+0.119, +0.178]	4.45	<0.001
C vs. C′	+0.019	[−0.001, +0.040]	0.82	0.070
C′ vs. D	+0.054	[+0.034, +0.074]	2.33	<0.001
CNN-MSE vs. C	−0.049	[−0.069, −0.028]	2.13	<0.001
CNN-NLL vs. CNN-MSE	+0.018	[−0.012, +0.047]	0.54	0.238
B vs. A	+0.075	[+0.046, +0.104]	2.29	<0.001

Key findings: at 200/5, D-DiffTS is statistically indistinguishable from C (p=0.403) but significantly exceeds D (d=1.58, p<0.001). At 500/10 and 1000/20, D-DiffTS significantly exceeds both C and D (all p<0.01). The frozen-encoder advantage is the largest (D-DiffTS vs. frozen C: d=2.50), confirming that learned negatives benefit most from preserved pre-trained features. The generator ablation shows that post-hoc guidance is essential (d=3.08, p<0.001): without it, learned negatives underperform analytical ones.

J Extended Scaling Analysis

J.1 NLL-Only Early Stopping

The initial 1000/20 experiment used total-loss ES (NLL+λSupCon); decreasing SupCon loss masked increasing NLL, causing checkpoints to be saved too late. Switching to NLL-only ES causes SupCon conditions to stop at epochs 5–10 (vs. running all 20), substantially improving checkpoint quality. B also improves under NLL-only ES (0.496→0.521 at 1000/20) due to better checkpoint selection.

Table 12: D-DiffTS pairwise comparisons of stealth-95 ROC-AUC. Conditions at 200/5 use 10 seeds; all others use 5 seeds. Δ: mean difference (A$-$B); CI: bootstrap 95% confidence interval; d: Cohen's d; p: bootstrap two-sided p-value.

Scale	Comparison	Δ AUC	95% CI	Cohen's d	p
200/5	D-DiffTS vs. C	-0.008	[-0.028, $+0.042$]	0.12	0.403 (n.s.)
200/5	D-DiffTS vs. D	$+0.065$	[$+0.034$, $+0.101$]	1.58	<0.001
500/10	D-DiffTS vs. C	$+0.059$	[$+0.010$, $+0.106$]	1.34	0.009
500/10	D-DiffTS vs. D	$+0.098$	[$+0.055$, $+0.139$]	2.58	<0.001
1000/20	D-DiffTS vs. C	$+0.068$	[$+0.028$, $+0.109$]	1.88	<0.001
1000/20	D-DiffTS vs. D	$+0.079$	[$+0.050$, $+0.112$]	2.78	<0.001
Frozen	D-DiffTS vs. C	$+0.164$	[$+0.089$, $+0.233$]	2.50	<0.001
Frozen	D-DiffTS vs. D	$+0.106$	[$+0.054$, $+0.161$]	2.16	<0.001
Gen. ablation	Full vs. no-guide	$+0.145$	[$+0.104$, $+0.190$]	3.08	<0.001
Degradation	500/10 vs. 1000/20	$+0.081$	[$+0.034$, $+0.126$]	1.94	<0.001
Degradation	200/5 vs. 1000/20	$+0.044$	[$+0.006$, $+0.085$]	1.00	0.011

J.2 Best-Epoch Analysis

Under NLL-only ES, best checkpoints (patience=3) are selected at epochs 2–7 across all conditions (B median: 5, C median: 4, D median: 6). These match the 200/5 epoch budget; the 500/10$\rightarrow$1000/20 degradation despite similar epoch counts suggests the degradation is data-dependent, not epoch-dependent.

J.3 Freezing: Mitigation Details

With 10-seed evaluation, frozen B=C=D=0.549$\pm$0.109—identical across conditions because NLL-based evaluation depends only on the frozen encoder, not the (differing) projection heads. The projection head is trained but never used during NLL scoring. Frozen AUC (0.549) exceeds zero-shot A (0.481) evaluated on the same synthetic_1k data, confirming that pre-trained features provide a detection floor above chance. However, the frozen AUC does not exceed the best unfrozen condition at 200/5 (C=0.629), indicating that the encoder's pre-trained representations alone are insufficient for strong detection without adaptation. Frozen-encoder F1 is low (0.152$\pm$0.077), reflecting the NLL scorer's inability to leverage the contrastive signal learned by the projection head. **Practitioner caveat**: the frozen-encoder protocol preserves pre-trained features but does *not* recover operational F1 under the benign-calibrated threshold. Deploying the frozen protocol requires either (a) an evaluation method that uses the trained projection head (e.g., contrastive-embedding distance), (b) a larger benign calibration set to reduce threshold variance, or (c) accepting the AUC floor as a safety net within a multi-layered IDS. **Note on 5-seed vs. 10-seed frozen results**: earlier 5-seed frozen experiments reported differentiated B/C/D values (0.606/0.582/0.573); the 10-seed results (B=C=D=0.549) are authoritative and supersede the 5-seed values throughout this paper.

J.4 Partial Freezing

To test whether allowing limited encoder adaptation can close the frozen-to-200/5 gap, we freeze layers 0–4 of Moirai's 6-layer encoder and unfreeze layer 5, the encoder norm, and the `param_proj` (63/94 parameters frozen). Partial-freeze C achieves AUC=0.590$\pm$0.106 and D achieves 0.595$\pm$0.110 (Table 3), marginally above the fully frozen results (C: +0.008, D: +0.022) and above the 200/5 baseline (C=0.563, D=0.479). The high variance persists ($\pm$0.106–0.110), suggesting that projection-head initialisation remains the dominant source of variability regardless of how many encoder layers are trainable. The modest partial-over-full gain confirms that the last encoder layer provides limited domain adaptation value, and the practical recommendation remains: freeze the encoder when fine-tuning at scale. A suggestive observation: under partial freezing, D (0.595) marginally exceeds C (0.590), reversing the 200/5 ranking (C=0.629 > D=0.556). Under full freezing, B=C=D=0.549 (10 seeds), so no condition differentiation is observable. The partial-freeze reversal is within noise ($\pm$0.106–0.110), but the pattern is consistent with stealth-controlled negatives providing more informative gradient signal when pre-trained features are partially preserved. If confirmed at larger scale or with reduced variance, this would reverse the paper's 200/5 conclusion about negative type.

J.5 Learning-Rate and LoRA Ablation

Table 13: Learning-rate and LoRA ablation at 1000/20 (stealth-95, 5 seeds, benign-calibrated threshold). Unfrozen baseline and LR/LoRA rows use 5 seeds; frozen row uses 10 seeds (B=C=D, identical; see Table 3). **Key findings**: (1) 10^{-5} partially mitigates degradation for C but not B or D. (2) 10^{-6} underfits across all conditions. (3) LoRA (rank=8) underperforms full freezing, suggesting low-rank adapters lack capacity for this task. (4) Full encoder freezing remains the most effective mitigation.

		AUC		F1		FPR	
Cond.	Configuration	mean	std	mean	std	mean	std
Unfrozen encoder (baseline, Table 3)							
B	lr=1e-4	.521	.024	.177	.031	.060	.049
C	lr=1e-4	.508	.041	.175	.048	.060	.049
D	lr=1e-4	.506	.029	.164	.059	.060	.049
Lower learning rate							
B	lr=1e-5	.480	.131	.152	.046	.060	.049
C	lr=1e-5	.557	.103	.243	.086	.100	.063
D	lr=1e-5	.476	.131	.177	.067	.080	.040
B	lr=1e-6	.465	.123	.138	.074	.020	.040
C	lr=1e-6	.471	.092	.087	.087	.080	.075
D	lr=1e-6	.465	.125	.138	.074	.020	.040
Parameter-efficient fine-tuning							
C	LoRA (r=8)	.484	.108	.234	.097	.100	.063
D	LoRA (r=8)	.484	.133	.214	.089	.100	.063
Frozen encoder (10 seeds, Table 3; B=C=D identical)							
B=C=D	Frozen	.549	.109	.152	.077	.070	.090

To test whether the 1000/20 degradation is explained by learning-rate misconfiguration (reviewer concern W3/Q1), we re-run conditions B, C, and D at lr$\in\{10^{-5}, 10^{-6}\}$ (5 seeds each). Results (Table 13): 10^{-5} partially mitigates degradation for C (0.557 vs. 0.508 at 10^{-4}) but not B (0.480) or D (0.476). At 10^{-6}, all conditions underfit (0.465–0.471), confirming a narrow LR window. LoRA (rank=8, α=16) targeting `q_proj`, `v_proj`, and `out_proj` in all encoder layers also underperforms full freezing (C: 0.484 vs. 0.549), suggesting low-rank adapters lack sufficient capacity. Together, these results rule out LR misconfiguration as the primary explanation for the scaling degradation.

J.6 L2-SP Regularisation Baseline

L2-SP [Li et al.(2018)Li, Grandvalet, and Davoine] penalises drift from pre-trained weights by adding $\lambda_{\text{L2-SP}} \sum_i \|w_i - w_i^0\|^2$ to the training loss, where w_i^0 are the pre-trained Moirai encoder parameters. We run conditions B, C, and D at 1000/20 with $\lambda_{\text{L2-SP}} = 0.01$ (5 seeds, NLL-only early stopping).

Table 14: L2-SP regularisation at 1000/20 (stealth-95). Unfrozen and frozen are 10-seed; L2-SP is 5-seed. Frozen B=C=D are identical (see text).

Condition	AUC (unfrozen)	AUC (L2-SP)	AUC (frozen)
B	$0.550_{\pm.088}$	$0.555_{\pm.141}$	
C	$0.581_{\pm.082}$	$0.555_{\pm.141}$	$0.549_{\pm.109}$
D	$0.557_{\pm.088}$	$0.555_{\pm.141}$	

At λ=0.01, L2-SP produces *identical* AUC across all three conditions (B=C=D=0.555±0.141), indicating the weight-drift penalty dominates the loss landscape and negates any negative-type effect. L2-SP AUC (0.555) is comparable to both unfrozen B (0.550) and frozen B=C=D (0.549, 10 seeds), confirming that the weight-drift penalty provides marginal benefit over full freezing at this regularisation strength. For C, L2-SP (0.555) actually *underperforms* unfrozen (0.581), suggesting the regularisation constrains the encoder's ability to benefit from SupCon gradients. The high variance (±0.141) reflects seed-dependent interactions between the L2-SP penalty and the SupCon contrastive signal.

J.7 Controlled Cross-Evaluation: Frozen and Unfrozen D-DiffTS

The main scaling table (Table 3) reports frozen D-DiffTS AUC=0.728 (5 seeds) using DiffTS-generated evaluation negatives, and unfrozen D-DiffTS AUC=0.576 (5 seeds) using the same. Because these evaluation negatives differ from the analytical negatives used for conditions B, C, and D, direct cross-condition comparison is unreliable. To resolve this, we decouple training data from evaluation data using a `-eval-synthetic-dir` argument that preserves training on DiffTS negatives while switching the evaluation set to the analytical negatives used for B/C/D.

Methodology. We run condition D at 1000/20, 5 seeds, with `-synthetic-dir data/synthetic_diffts` (training) and `-eval-synthetic-dir data/synthetic` (evaluation), under both frozen and unfrozen encoder settings. The evaluation set is identical to that used for frozen B=C=D=0.549 (10 seeds) and for unfrozen B/C/D.

Results.

Table 15: Controlled cross-evaluation of D-DiffTS (train on DiffTS negatives, evaluate on analytical negatives; stealth-95, 5 seeds). For comparison, B/C/D results use 10 seeds on the same analytical evaluation set.

Condition	Encoder	AUC	F1
D-DiffTS	Unfrozen	$0.463_{\pm.073}$	$0.171_{\pm.071}$
D-DiffTS	Frozen	$0.464_{\pm.060}$	$0.183_{\pm.074}$
B (analytical eval)	Unfrozen	$0.550_{\pm.088}$	$0.192_{\pm.071}$
C (analytical eval)	Unfrozen	$0.581_{\pm.082}$	$0.206_{\pm.080}$
D (analytical eval)	Unfrozen	$0.557_{\pm.088}$	$0.167_{\pm.065}$
Frozen B=C=D	Frozen	$0.549_{\pm.109}$	$0.152_{\pm.077}$
Zero-shot A	—	0.481	—

Interpretation. Both frozen (0.464) and unfrozen (0.463) D-DiffTS fall below zero-shot A (0.481) when evaluated on analytical negatives. Unfrozen D-DiffTS is decisively below unfrozen B (0.550), C (0.581), and D (0.557). This confirms two artifacts reported elsewhere in the paper: (1) the frozen 0.728 was an evaluation-distribution artifact: the frozen encoder trivially separates DiffTS negatives (different feature-space region) rather than benefiting from D-DiffTS training; (2) the unfrozen D-DiffTS apparent advantage at 1000/20 (0.576 vs. C 0.581 on their respective eval sets) is also an evaluation-distribution effect—not a genuine generalisation advantage of learned negatives. The greater weight drift in D-DiffTS (Table 16) therefore does not produce more useful representations for detecting analytical-negative attacks; it specialises the encoder toward the DiffTS feature space.

J.8 Diffusion-TS Training Details

We train a Diffusion-TS [Zhang et al.(2024)] model on the CICIoT2023 benign training split to generate learned negatives.

Architecture. The model uses the decomposition-aware Diffusion-TS architecture: a 3-layer Transformer encoder and 6-layer Transformer decoder with d_{model}=256, input sequence length 128, and 12 features. The diffusion process uses a cosine noise schedule with 1000 timesteps.

Training. We train for 300 epochs with Adam (η=10^{-4}), batch size 32, on the benign training windows. The checkpoint with lowest validation reconstruction loss is selected (best loss: 0.238 at epoch 277). Training time: $\approx$12 hours on CPU (Apple M-series). The training seed is fixed (42) for reproducibility.

Sampling. At inference, we use 50 DDIM denoising steps (vs. 1000 full steps), providing a 20$\times$ speedup with negligible quality loss. Samples are generated in batches of 50; generating 1000 benign base samples takes $\approx$25 minutes.

Post-hoc statistical guidance. After generating each base sample, we inject analytical attack perturbations (Eqs. 2–5) identically to the analytical pipeline. Each perturbed sample is then rescaled to match target statistics: the mean is set to $\mu(\mathbf{X}_b)$ (the corresponding real benign sample) and the standard deviation is set to $\sigma(\mathbf{X}_b) \cdot (1 + (1{-}s) \cdot 0.5)$, where s is the stealth level. This ensures the learned negatives match the intended stealth profile while preserving the temporal structure from the generative model.

Generator ablation design. The D-DiffTS pipeline has three components: (1) learned base samples from Diffusion-TS, (2) analytical attack perturbations, and (3) post-hoc statistical guidance. To isolate which component drives any performance difference, we evaluate D-DiffTS-noguide (components 1+2 only, skipping the rescaling step) alongside the full D-DiffTS (components 1+2+3) and D-analytical (real benign base + component 2 only).

J.9 MMD Between Analytical and DiffTS Distributions

To quantify the distributional difference between analytical (D) and learned (D-DiffTS) negatives, we compute Maximum Mean Discrepancy (MMD) [Gretton et al.(2012)Gretton, Borgwardt, Rasch, Schölkopf, and Smola] with an RBF kernel (bandwidth via median heuristic) between the two distributions at each attack type and stealth level (200 samples each, flattened to $\mathbb{R}^{128\times 12}$).

Aggregate. The overall MMD=0.059 is statistically significant (p=0.002, 500 permutations), confirming that analytical and DiffTS-generated negatives occupy measurably different regions of the input space.

Per-attack-type. MMD is highest for protocol anomaly (stealth-85: 0.047, p=0.01) and slow exfiltration (stealth-85: 0.046, p=0.008), where the Diffusion-TS generator produces the most distributional divergence from the analytical pathway. At stealth-95, protocol anomaly (MMD=0.017, p=0.31) and slow exfiltration (MMD=0.022, p=0.25) are no longer significant, consistent with both generators converging toward the benign distribution at high stealth. Beacon and LotL mimicry show intermediate divergence across stealth levels, with beacon consistently significant ($p \leq 0.04$).

Interpretation. The statistically significant distributional difference at lower stealth levels suggests that DiffTS negatives provide genuinely different training signal compared to analytical negatives, supporting the interpretation that the D-DiffTS advantage at larger scales reflects distributional richness rather than a confound. The convergence at stealth-95 is expected: higher stealth forces both generation pathways toward the benign manifold, reducing distributional divergence.

K Extended Discussion

K.1 Why Does Base Moirai Underperform?

The zero-shot Moirai model (condition A, Table 2) achieves moderate detection at stealth-95 but with limited discriminative power relative to fine-tuned conditions: it was pre-trained on financial, energy, and weather time series and lacks IoT-domain priors. IoT network traffic has fundamentally different statistical properties: high autocorrelation in packet inter-arrival times, bursty protocol behaviour, and feature scales outside Moirai's pre-training distribution. Fine-tuning with the NLL+SupCon objective and stealth-controlled negatives (condition D vs A, Table 2) improves AUC and reduces FPR, confirming that domain adaptation is the primary bottleneck.

K.2 Gaussian-Noise vs Stealth-Controlled Negative Mechanism

Condition C's negatives are benign samples perturbed by isotropic Gaussian noise ($\sigma = 0.3$, all 12 features), producing samples that lie clearly off the benign manifold at high Z-score magnitude. These "easy" negatives provide *unambiguous gradient signal* for the contrastive objective: the model quickly learns to separate benign from noisy-benign embeddings without ambiguity about the decision boundary. Synthetic stealth-controlled negatives (stealth 85–95%), by contrast, are deliberately positioned near the benign manifold; at limited training scale this proximity increases embedding ambiguity and may prevent the projection head from establishing a reliable boundary within the allotted epochs. The C′ > D comparison (+0.054 AUC, 0.610 vs. 0.556; p<0.001) further shows that even within stealth-controlled negatives, the 1:12 imbalance compounds the problem: the large excess of stealth-controlled negative training samples overwhelms the benign signal, shifting the decision boundary to favour recall over precision. The scaling experiment (Table 3) tests this directly. Under total-loss ES at 1000/20, C degrades to AUC=0.520 (−0.043 from 200/5), C′ to 0.500 (−0.034), B to 0.496 (+0.015), and D to 0.501 (+0.022). Consistent with this interpretation, easy negatives (C) degraded most (−0.043); B and D actually improved slightly at 1000/20 under total-loss ES, suggesting their 200/5 baselines were under-trained.

With NLL-only early stopping (conditions stop at epochs 5–10), C=0.508 and D=0.506 (gap=0.003, 95% bootstrap CI $[-0.014, +0.020]$, p=0.39, n.s.; 5 seeds). Note: the main text reports p=0.216 for the same C>D comparison at 1000/20 (Table 3); these differ because the main-text figure uses 10 seeds and the same NLL-only early-stopping criterion applied to the full 10-seed run, while this appendix section reflects an earlier 5-seed analysis under the same criterion—the two estimates are consistent in direction and n.s. status. The C>D gap at 200/5 was a small-scale training artefact for *analytical* negatives: easy off-manifold negatives provide clearer gradient signal when training budget is limited. D-DiffTS learned negatives close this gap at 200/5 (0.621 vs. C 0.629, p=0.403; Table 12) and substantially exceed C at larger scales, confirming that the gap reflects analytical-perturbation quality, not stealth-controlled negatives in principle.

K.3 Catastrophic Forgetting Mechanism

The degradation from 200/5 to 1000/20 (Table 3) is explained by the encoder-freezing experiment. With 10 seeds, freezing Moirai's pre-trained weights and training only the projection head at 1000/20 yields B=C=D=0.549±0.109—identical across conditions because NLL-based evaluation scores from the frozen encoder directly and the projection head is not used in scoring. Extended fine-tuning catastrophically overwrites the pre-trained time-series features that provide discriminative power for anomaly detection. At 200/5, the small training budget limits weight drift, so the pre-trained features survive; at 1000/20, the longer training progressively destroys them, even though NLL-only ES selects checkpoints at epochs 2–7 (matching the 200/5 budget). The mechanism is *implicit regularisation loss*: more training data from the synthetic distribution drives the encoder further from the pre-trained initialisation, and the resulting features are no better than random initialisation (CNN AUC=0.513≈unfrozen Moirai at 1000/20: B=0.550, C=0.581, D=0.557).

The frozen AUC of 0.549 exceeds zero-shot A (0.481) evaluated on the same data, confirming that pre-trained features provide a detection floor above chance. However, frozen AUC does not exceed the best unfrozen condition at 200/5 (C=0.629), indicating that the encoder's pre-trained representations alone are insufficient without adaptation. **Frozen D-DiffTS** achieves AUC=0.728±0.080 (5 seeds) on DiffTS-generated evaluation data, but a controlled cross-evaluation on the same analytical negatives as B/C/D yields AUC=0.464—below zero-shot A (0.481) and far below frozen B=C=D=0.549 (Appendix J.7). The 0.728 figure is an evaluation-distribution artifact and should not be interpreted as evidence of D-DiffTS training benefit. Partial freezing (layers 0–4 frozen, layer 5 unfrozen) yields a modest further gain (partial C=0.590, D=0.595), and the high variance persists (±0.106–0.110).

K.4 Direct Measurement of Encoder Weight Drift

To provide direct evidence of catastrophic forgetting beyond the indirect signal from encoder-freezing experiments, we measure per-layer cosine distance and Frobenius norm ratio between pre-trained and fine-tuned Moirai encoder weights across all three negative conditions at each training scale.

Table 16: Encoder weight drift and representational similarity between pre-trained and fine-tuned Moirai encoder, measured at the checkpoint selected by early stopping. Cosine distance and Frobenius ratio are computed per-layer then averaged. Linear CKA is computed on mean-pooled encoder representations for a 50-sample benign probe set (CKA=1 means identical representations; lower=more change). "Stop Ep." is the epoch at which early stopping triggered (patience=3).

Condition	Scale	Mean Cos. Dist.	Mean Frob. Ratio	CKA	Stop Ep.
C	200/5	0.000143	0.013	0.426	5
D	200/5	0.000135	0.012	0.436	5
D-DiffTS	200/5	0.000164	0.014	0.466	5
C	500/10	0.000395	0.021	0.443	9
D	500/10	0.000373	0.020	0.400	10
D-DiffTS	500/10	0.000427	0.021	0.343	10
C	1000/20	0.000432	0.022	0.373	6
D	1000/20	0.000556	0.024	0.381	9
D-DiffTS	1000/20	0.001950	0.041	0.183	14

Three patterns emerge. First, drift increases monotonically with scale for all conditions, consistent with the catastrophic forgetting hypothesis. Second, at 1000/20, D-DiffTS shows the highest weight drift (cosine distance=0.00195, +352% vs. C) and the lowest CKA (0.183 vs. C 0.373), indicating the most representational change. On its own DiffTS-generated evaluation set, D-DiffTS achieves AUC=0.576 (5 seeds)—but this comparison uses a different evaluation distribution than C. A controlled cross-evaluation (unfrozen D-DiffTS evaluated on the same analytical negatives as B/C/D) yields AUC=0.463±0.073 (5 seeds, stealth-95), below zero-shot A (0.481) and below all unfrozen B/C/D conditions. This indicates that greater weight drift in D-DiffTS does *not* produce more useful representations for detecting analytical perturbations; the apparent within-distribution advantage reflects evaluation-set mismatch. Third, when the encoder is frozen, D-DiffTS achieves AUC=0.728 (5 seeds) on DiffTS-generated evaluation data—but controlled cross-evaluation on analytical negatives yields AUC=0.464±0.060, confirming this was an evaluation-distribution artifact (Appendix J.7).

K.5 CNN Baseline: Detailed Analysis

Two interpretations are consistent with the CNN outperformance. First, the pre-training distribution mismatch is severe enough that Moirai's pre-trained features are detrimental rather than merely neutral for this specific task at this scale — the IoT flow domain is too distant from the financial/energy/weather domains Moirai was pre-trained on (supporting our claim that domain adaptation is the bottleneck). Second, Moirai's large parameter space introduces optimisation instability at small training scale, while the CNN's simpler architecture converges more reliably—the higher Moirai variance (±0.022 vs. CNN ±0.023) supports this interpretation (note: at 10 seeds, variance has stabilised).

Loss-function confound resolved. To rule out the possibility that the CNN advantage stems from MSE being an easier optimisation target than distributional NLL, we train a CNN variant with a Gaussian NLL head (CNN-NLL): predict μ and $\log \sigma$ per feature, with loss $\frac{1}{2}(\log \sigma + (x-\mu)^2/\sigma^2)$. CNN-NLL (AUC=0.598±0.041) and CNN-MSE (AUC=0.580±0.023) are not significantly different (p=0.24, d=0.54, bootstrap 10,000 resamples). The loss function is *not* the explanation for CNN outperforming Moirai.

Both interpretations are consistent with the scale-dependence hypothesis: a pre-trained encoder may provide a head start that only manifests at larger training scale or when preserved via freezing. The scaling experiment fills the CNN row in Table 3: at 1000 samples / 20 epochs, CNN achieves AUC=0.513±0.029 (stealth-95), degrading from 0.597 ($\Delta=-0.084$)—while B degrades from 0.481 to 0.521 ($\Delta=+0.040$), C from 0.563 to 0.508 ($\Delta=-0.055$), and D from 0.479 to 0.506 ($\Delta=+0.027$). (All baselines here are 5-seed values from Table 3.) CNN's degradation at scale is the most severe, consistent with the absence of pre-trained features to fall back on.

K.6 SupCon Gradient Flow Analysis

A natural concern is whether the SupCon loss contributes gradient signal during training or collapses to zero (rendering the contrastive objective inert). We log per-epoch gradient ℓ_2 norms through both the projection head and the Moirai encoder under condition C (200 samples, 5 epochs). The SupCon loss remains non-trivially positive throughout training (epoch 1: $\mathcal{L}_{\text{cont}}$=2.15; epoch 5: 1.98), and gradient norms on the encoder range from 53.6 (epoch 1) to 9.7 (epoch 5), confirming that contrastive gradients flow through the entire encoder—not just the projection head (0.55 to 0.01). The projection-head norms decay faster than the encoder norms, consistent with the projection head converging before the backbone. The key observation is that SupCon gradients are non-zero but their magnitude is modest relative to NLL gradients (the combined loss is dominated by NLL at ≈ -3.5 vs. SupCon at $\approx +2.0$), supporting the NLL-dominance hypothesis at this training scale.

K.7 Alternative Contrastive Frameworks

At this scale and domain, NLL-only fine-tuning (B, AUC=0.556) performs identically to the full system (D, AUC=0.556), and SupCon's gradient magnitude is modest relative to NLL (Appendix K.6). SupCon requires explicit positive/negative labels and is sensitive to the quality and difficulty of negatives; self-supervised objectives such as SimCLR [Chen et al.(2020)Chen, Kornblith, Norouzi, and Hinton] or BYOL [Grill et al.(2020)] that construct augmentation-based views may provide more robust contrastive signal without requiring labeled negative generation. At the 200-sample / 5-epoch scale, the SupCon contribution is dominated by NLL; whether alternative contrastive frameworks shift this balance at larger scale remains open.

K.8 Ensemble Degradation with Stealth Level

Under the benign-calibrated threshold protocol (Table 1), the Ensemble's F1 drops from 0.221 (stealth-85) to 0.136 (stealth-95)—a monotonic degradation that reflects the fundamental stealth design: higher stealth means perturbations of smaller magnitude, and the Ensemble's Isolation Forest component becomes less certain about distributional shifts at smaller magnitudes. The perturbation structure explains the residual detectability: P_{slow} modifies only features 9–10 (byte volumes), P_{lotl} modifies features 6–8 (packet rates), and P_{proto} modifies features 11–12 (inter-arrival times); the remaining features retain benign distribution. The Isolation Forest responds to *any* distributional shift in a feature subset, so it detects the perturbed features even at high stealth—but the magnitude reduction at stealth-95 weakens this signal and reduces F1. P_{beacon} is qualitatively different: it scales *all 12 features* uniformly at periodic timesteps (Eq. 4). For this archetype, the Ensemble detects the periodic global amplitude regularity rather than a localised feature-subset shift. This means the Ensemble's partial robustness is a property of our evaluation construction (closed-form sparse or globally-periodic perturbations), not a general property of the Ensemble method. Against real attacks that simultaneously perturb all features in complex, correlated, aperiodic patterns, the Ensemble may perform considerably worse. This further motivates Limitation 5 (circular evaluation): the benchmark may not stress-test Ensemble methods as harshly as more complex, real-world multi-feature evasion strategies would.

K.9 Protocol-Constrained Generation Algorithm

The constraint-retry loop works as follows. Given a benign sequence $\mathbf{X}_b$, attack pattern P_k, stealth level s, and protocol specification π, the generator applies the closed-form archetype perturbation $\mathbf{X}_a \leftarrow P_k(\mathbf{X}_b, s)$, then checks the protocol constraint $\mathcal{C}_\pi(\mathbf{X}_a)$. If the constraint is violated, the stealth level is relaxed by $s \leftarrow \min(s + 0.01,\ 1.0)$ and the perturbation is re-applied, up to R=3 retries. In practice, >99% of samples pass on the first attempt at stealth ≤ 0.95.

L Leave-One-Out Generalization Details

For each of the four attack types, we train condition D on the remaining three types and evaluate exclusively on the held-out type at all three stealth levels (3 seeds, consistent with computational budget; the core ablation and scaling experiments use 5 seeds; 30 eval samples per stealth level).

Table 17: Leave-one-out generalization (stealth-95 F1, mean±std across 3 seeds, benign-calibrated threshold). Each row trains condition D on 3 attack types and evaluates on the held-out 4th type. Δ = HNIDS F1 − zero-shot F1. All four folds show positive transfer under calibrated evaluation.

Held-out Attack Type	Zero-shot F1	HNIDS F1	Δ
Slow Exfiltration	$0.144_{\pm.126}$	$0.295_{\pm.163}$	$+0.151$
LotL Mimicry	$0.114_{\pm.089}$	$0.367_{\pm.171}$	$+0.253$
C2 Beacon	$0.148_{\pm.110}$	$0.355_{\pm.050}$	$+0.206$
Protocol Anomaly	$0.148_{\pm.110}$	$0.283_{\pm.128}$	$+0.135$
Mean	0.139	0.325	$+0.186$

The largest gain is for `lotl_mimicry` (+0.253±0.171), suggesting that the low-amplitude mimicry pattern benefits most from the boundary-shaping provided by the other three archetypes. `beacon` (+0.206±0.050) and `slow_exfiltration` (+0.151±0.163) also show robust positive transfer. The high standard deviations (±0.05–0.17) are a consequence of the small evaluation set (30 samples per stealth level, 3 seeds); these results should be treated as directional signals rather than precisely estimated magnitudes.

M Extended N-BaIoT Analysis

M.1 Zero-Shot Failure Analysis

Table 8 (Appendix D) reports per-attack F1, FPR, and ROC-AUC across 10 Mirai and BASHLITE botnet variants (3 seeds, Danmini doorbell device). ROC-AUC values range from 0.013 (BASHLITE Combo) to 0.284 (Mirai Ack), all well below 0.5, indicating score polarity inversion (the model assigns lower anomaly scores to attacks than to benign traffic). FPR = 1.00 across all variants regardless of threshold placement.

This outcome is *expected and informative*: it demonstrates that the detector does not exploit trivial statistical shortcuts, and directly addresses the circular evaluation concern (Limitation 5, Section 6). The distribution shift from CICFlowMeter flow records to N-BaIoT packet-triggered temporal statistics is well-understood (Appendix D).

M.2 Preprocessing Details

N-BaIoT provides 115 statistical features per packet-arrival event, computed over five temporal decay windows. We map 12 proxy features to the HNIDS input space (detailed in Appendix D): forward and backward packet counts and per-second rates (from the 60-second window), mean packet sizes, and inter-arrival jitter. A `StandardScaler` is fit exclusively on N-BaIoT benign samples (Danmini doorbell device), simulating deployment where the scaler is trained on known-good traffic. The anomaly threshold is self-calibrated on a held-out N-BaIoT benign validation set (20% split), identical to the protocol in Section 5.3.

M.3 Recipe Generalisation Interpretation

The N-BaIoT results demonstrate recipe generalisation: NLL+SupCon transfers cross-domain. On device 1 (310 training windows), Gaussian negatives (AUC=0.928) significantly outperform D-DiffTS (AUC=0.660±0.427)—the training set is too small for DiffTS to learn a robust benign distribution. On device 4 (1,096 training windows), D-DiffTS achieves AUC=0.939±0.014, slightly exceeding Gaussian (0.929±0.012), confirming the $\geq$500-sample requirement and demonstrating that learned negatives match or exceed Gaussian when data is adequate. Condition C is domain-agnostic; D requires domain-specific archetypes.

N Baseline Hyperparameter Tuning

To address the concern that baselines use published defaults while HNIDS benefits from extensive tuning, we conduct a grid search over each deep-learning baseline (3 seeds, calibration AUC on 20% held-out split). Table 18 reports the grid, best configuration, and test-set performance.

Grid definitions. USAD: $z_{\text{dim}} \in \{20, 40, 80\}$, $\alpha \in \{0.3, 0.5, 0.7\}$, epochs $\in \{50, 100\}$ (18 configs). TranAD: $d_{\text{model}} \in \{128, 256\}$, layers $\in \{2, 4\}$, epochs $\in \{50, 100\}$ (8 configs). Anomaly Transformer: $d_{\text{model}} \in \{128, 256, 512\}$, epochs $\in \{50, 100\}$ (6 configs). PatchTST: patch size $\in \{8, 16, 32\}$, $d_{\text{model}} \in \{128, 256\}$ (6 configs).

The ranking is unchanged after tuning: USAD and Anomaly Transformer remain the strongest baselines, and the best tuned AUC (USAD, 0.565) remains 0.064 below HNIDS condition C (0.629). USAD and Anomaly Transformer benefit marginally from tuning (+0.017 and +0.023 AUC respectively), while TranAD and PatchTST show negligible change (−0.016 and −0.005). The best-tuned USAD configuration (z_{dim}=40, α=0.7) increases the reconstruction-loss weighting, consistent with the anomaly-detection objective.

Table 18: Baseline hyperparameter tuning (stealth-95, 200 eval samples, 3 seeds). "Default AUC" uses published hyperparameters (10 seeds, from Table 1); "Tuned AUC" uses the best grid-search configuration. Tuning does not change the ranking: no baseline exceeds USAD's tuned AUC of 0.565, which remains below HNIDS C (0.629).

Method	Configs	Best Config	Default AUC	Tuned AUC
USAD	18	z=40, α=0.7, ep=100	0.548±0.047	0.565±0.024
TranAD	8	d=256, layers=2, ep=100	0.517±0.057	0.501±0.026
Anom. Trans.	6	d=128, ep=50	0.519±0.039	0.542±0.026
PatchTST	6	patch=32, d=256	0.521±0.054	0.516±0.030

N.1 Supervised Baseline Classifiers

To isolate the effect of labeled attack data from the temporal-modeling and pre-trained feature contributions of HNIDS, we train three supervised classifiers on the same 72-dimensional statistical features (12 features × 6 statistics: mean, variance, skewness, kurtosis, min, max) derived from the same stealth-controlled negative windows used by condition D, evaluated on the same stealth-95 test set as the main results.

Feature extraction. Each 128-timestep window is summarised by computing 6 aggregate statistics per feature, yielding a 72-dim feature vector. This is the same feature set used by the statistical baselines in Table 1.

Classifiers. We train three classifiers:

- **LogReg**: sklearn `LogisticRegression` with `class_weight='balanced'`, preceded by `StandardScaler`.
- **XGBoost**: `GradientBoostingClassifier` (sklearn implementation; 100 estimators, max depth 3).
- **MLP**: sklearn `MLPClassifier` with two hidden layers (128, 32), early stopping on a 10% validation split.

Evaluation protocol. To prevent data leakage, we construct a strict held-out split: the available stealth-controlled negatives at each stealth level are divided 80/20 into training and evaluation sets, stratified by attack type. The 80% training split is used to fit the classifier; the 20% evaluation split (combined with a benign evaluation set of equal size) is used for AUC and F1 computation. Thresholds are calibrated to the 95th percentile of benign-only training scores (identical calibration protocol to HNIDS). We repeat for 10 seeds; each seed determines the train/eval split.

Results.

Table 19: Supervised classifiers on 72-dim statistical features vs. HNIDS (stealth-95, benign-calibrated threshold, 10 seeds). LogReg achieves AUC=0.630, matching HNIDS condition C (0.629).

Classifier	AUC	F1	FPR
LogReg	$0.630_{\pm.082}$	$0.312_{\pm.154}$	$0.087_{\pm.098}$
XGBoost	$0.570_{\pm.107}$	$0.238_{\pm.140}$	$0.038_{\pm.080}$
MLP	$0.453_{\pm.110}$	$0.086_{\pm.076}$	$0.113_{\pm.130}$
HNIDS B	$0.550_{\pm.088}$	$0.192_{\pm.071}$	—
HNIDS C	$0.581_{\pm.082}$	$0.206_{\pm.080}$	—
HNIDS D	$0.557_{\pm.088}$	$0.167_{\pm.065}$	—

Interpretation. LogReg (AUC=0.630±0.082) matches HNIDS condition C (0.629±0.022) and exceeds B and D at 1000/20. XGBoost (0.570±0.107) falls between B and C. MLP underperforms all HNIDS conditions (0.453±0.110). These results indicate that *labeled attack data* is the primary driver of the AUC improvement over zero-shot A (0.481): a linear classifier on hand-crafted features achieves the same detection rate as a fine-tuned foundation model with SupCon training. HNIDS's residual value over LogReg is: (1) lower FPR under calibrated thresholds (C: 0.068 vs. LogReg: 0.087, from Table 1), (2) the sequential encoder structure that enables the frozen-encoder forgetting diagnosis presented in this paper, and (3) a modular architecture that can incorporate improved negative-generation pipelines without retraining the feature extractor.

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